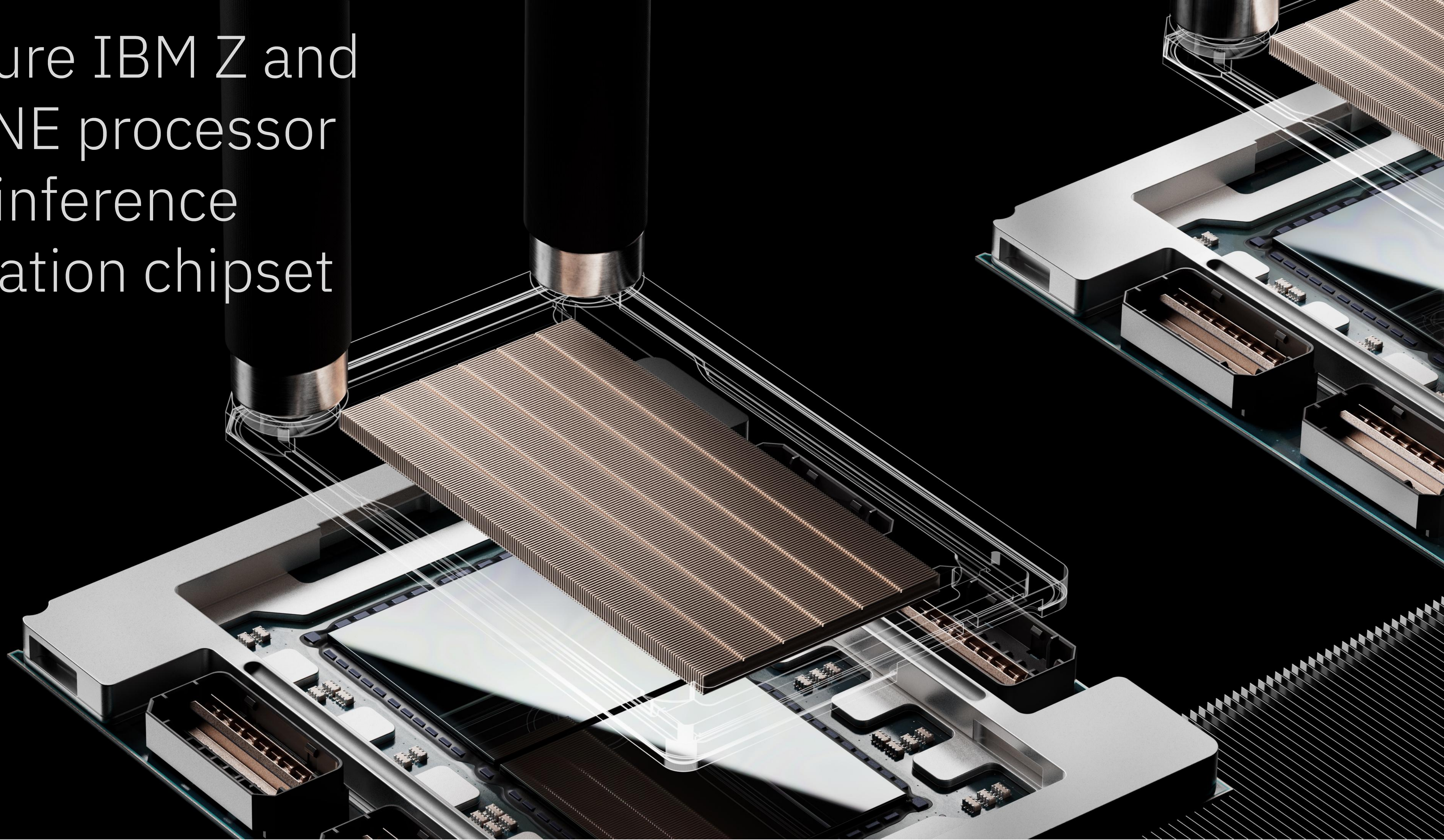


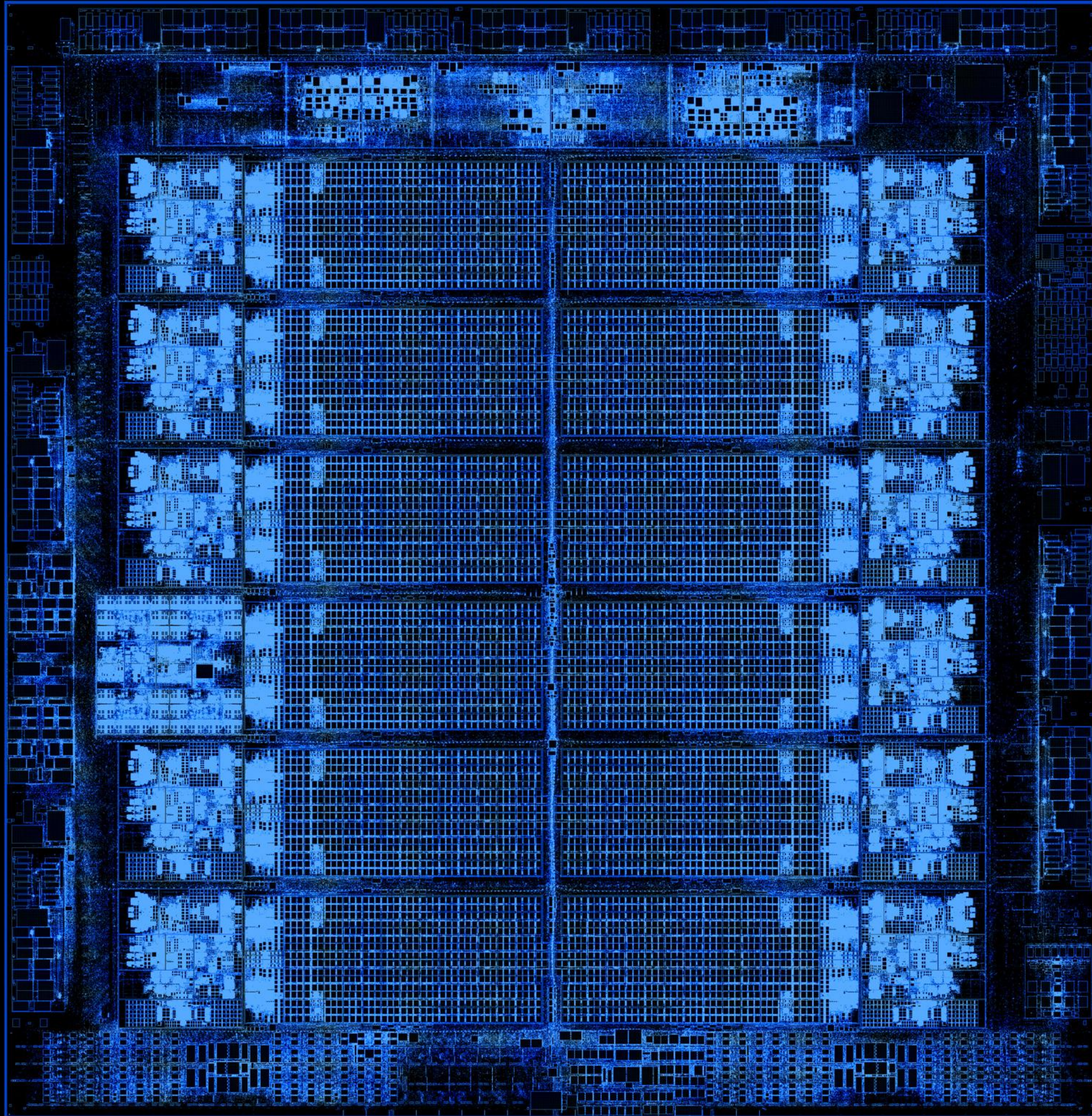
# The future IBM Z and LinuxONE processor and AI inference acceleration chipset



Christian Zoellin  
IBM Distinguished Engineer

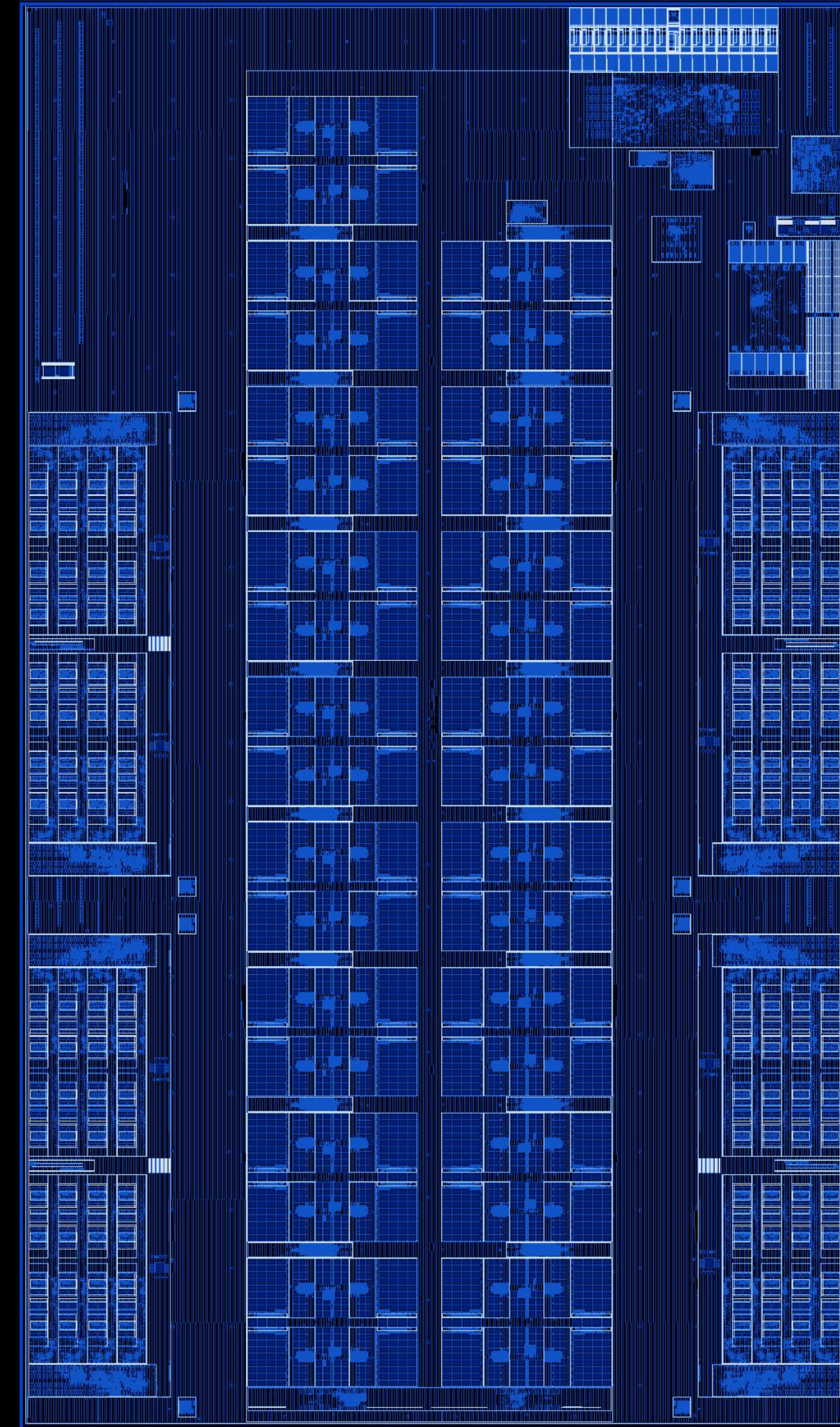
*Statements regarding IBM's future direction and intent are subject to change or withdrawal without notice and represent goals and objectives only.*





Dual-ISA Core  
z/Architecture  
+  
Arm-v9.3

# 2nd Generation AI Accelerator



Fit for →  
purpose



Designed for mission-critical  
enterprise workloads

Data



serving

Designed for mission-critical  
enterprise workloads

# Transaction



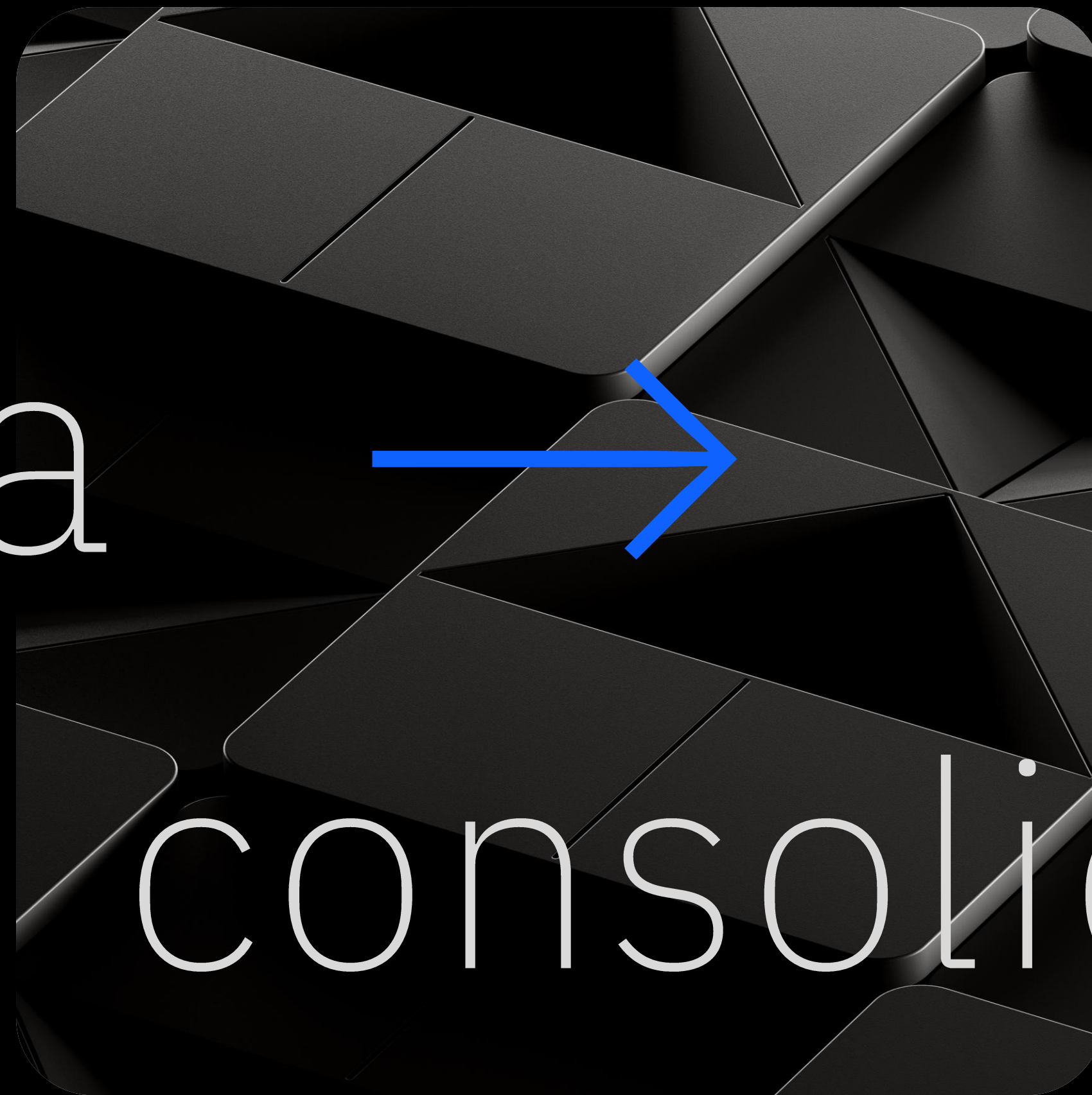
# serving

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enterprise workloads

Workload

d

consolidation



# 70%

of the world's  
transactions  
by value

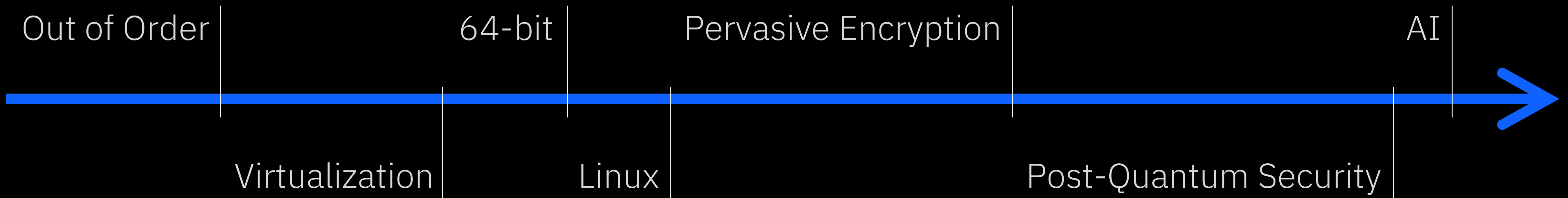
# 10000

of MongoDB  
instances in  
one mainframe

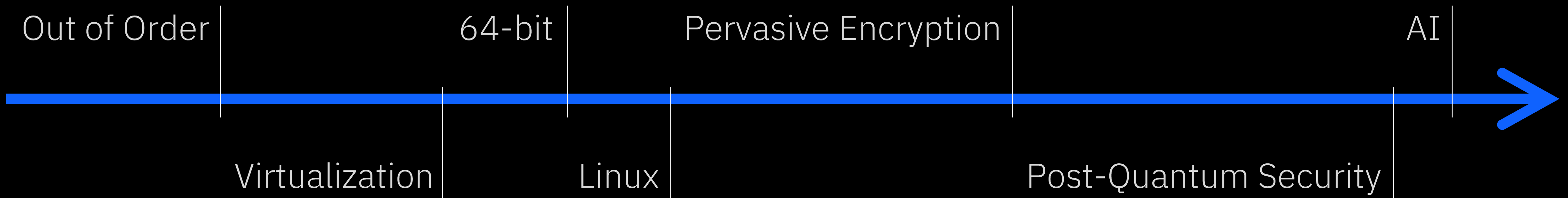
Reliability  
Security  
Scalability



IBM Z has a rich history  
because of relentless  
innovation



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# The Processor

5.7+ GHz, 2nm

## Core

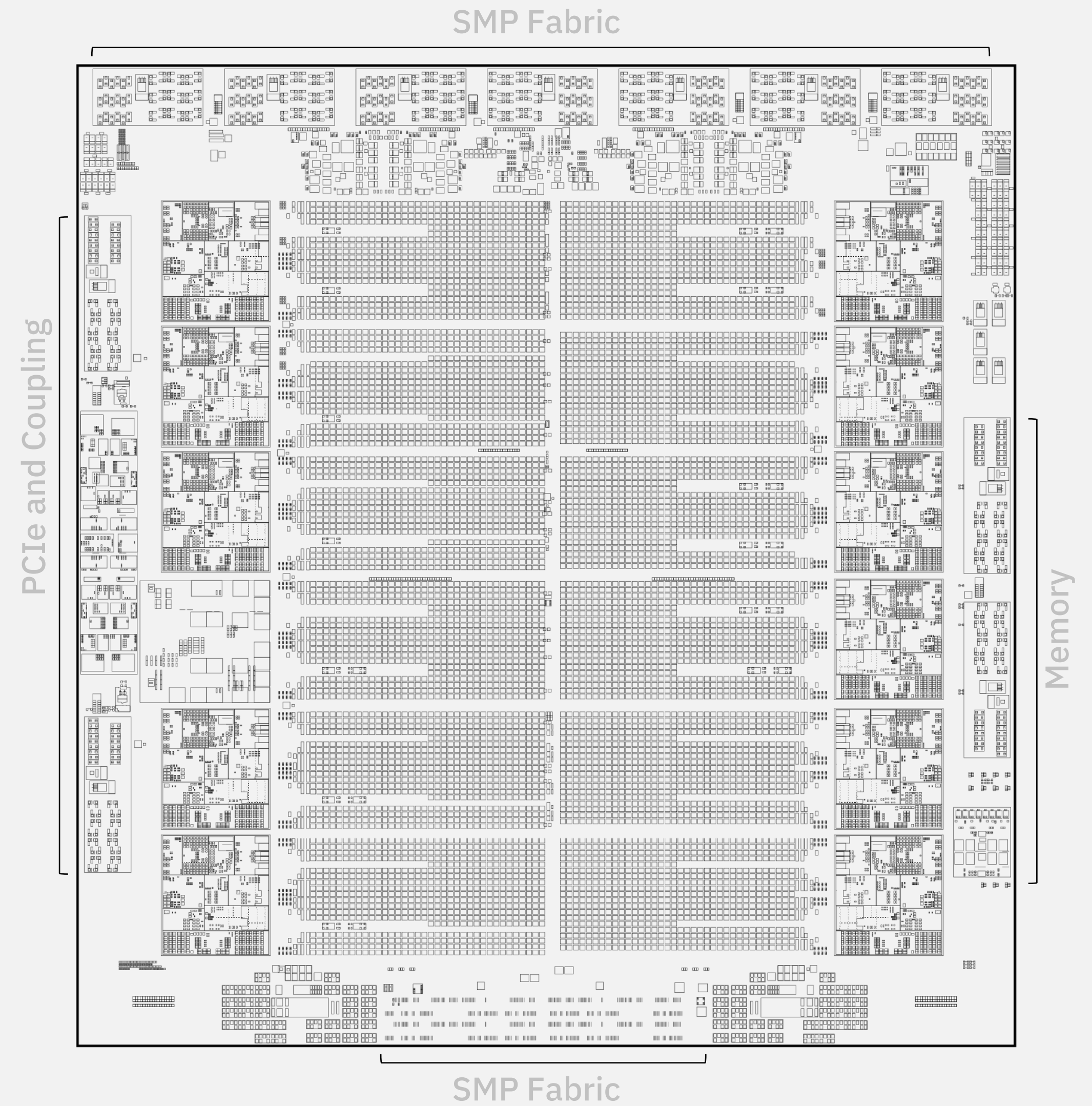
- 11 cores
- SMT-2
- Native z/Architecture and AArch64 execution

## L2 Cache

- 36MB private L2 caches
- 432MB Virtual-L3 and 3.5GB virtual-L4

## Acceleration

- Dedicated on-chip DPU for I/O acceleration
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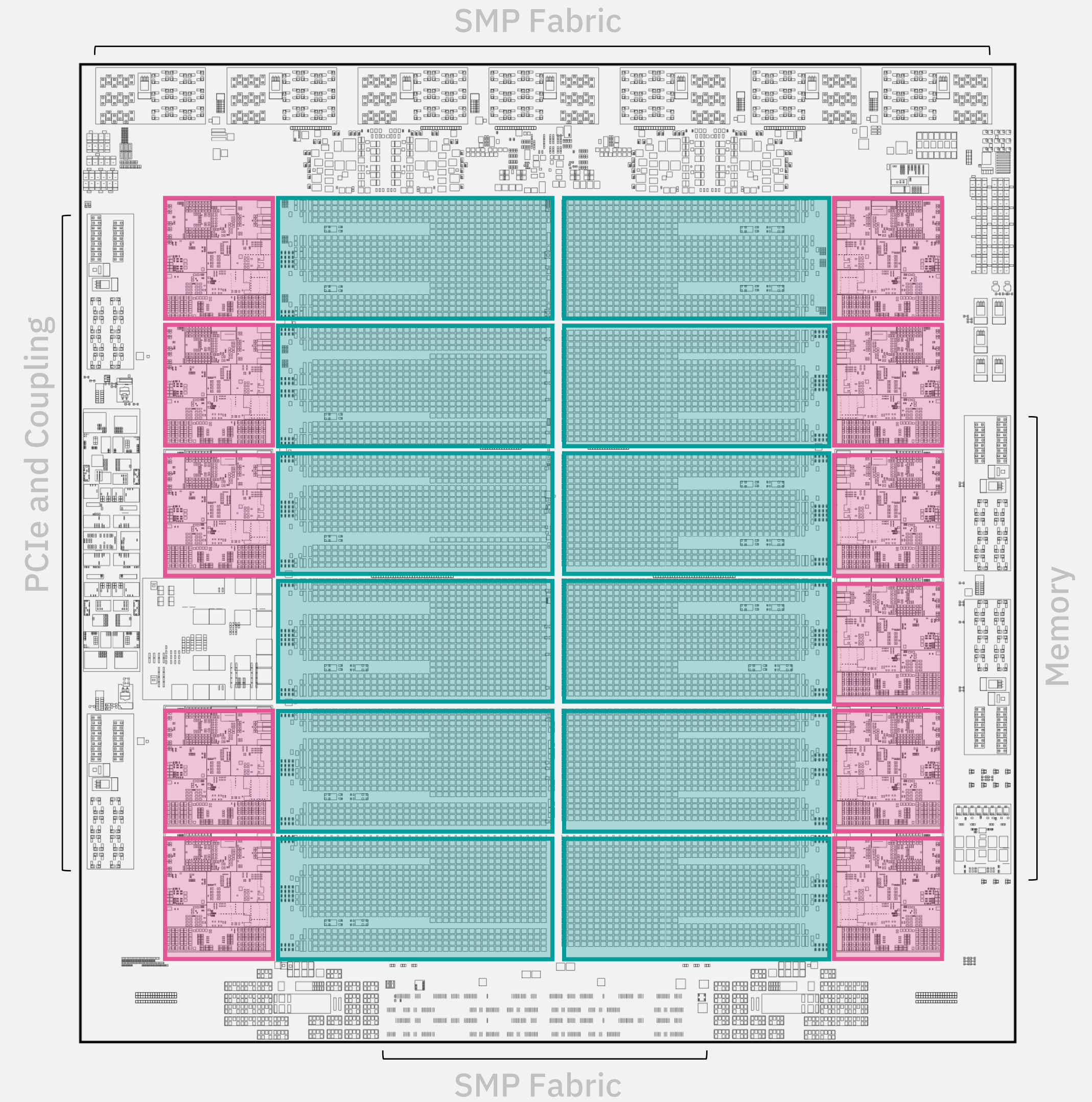
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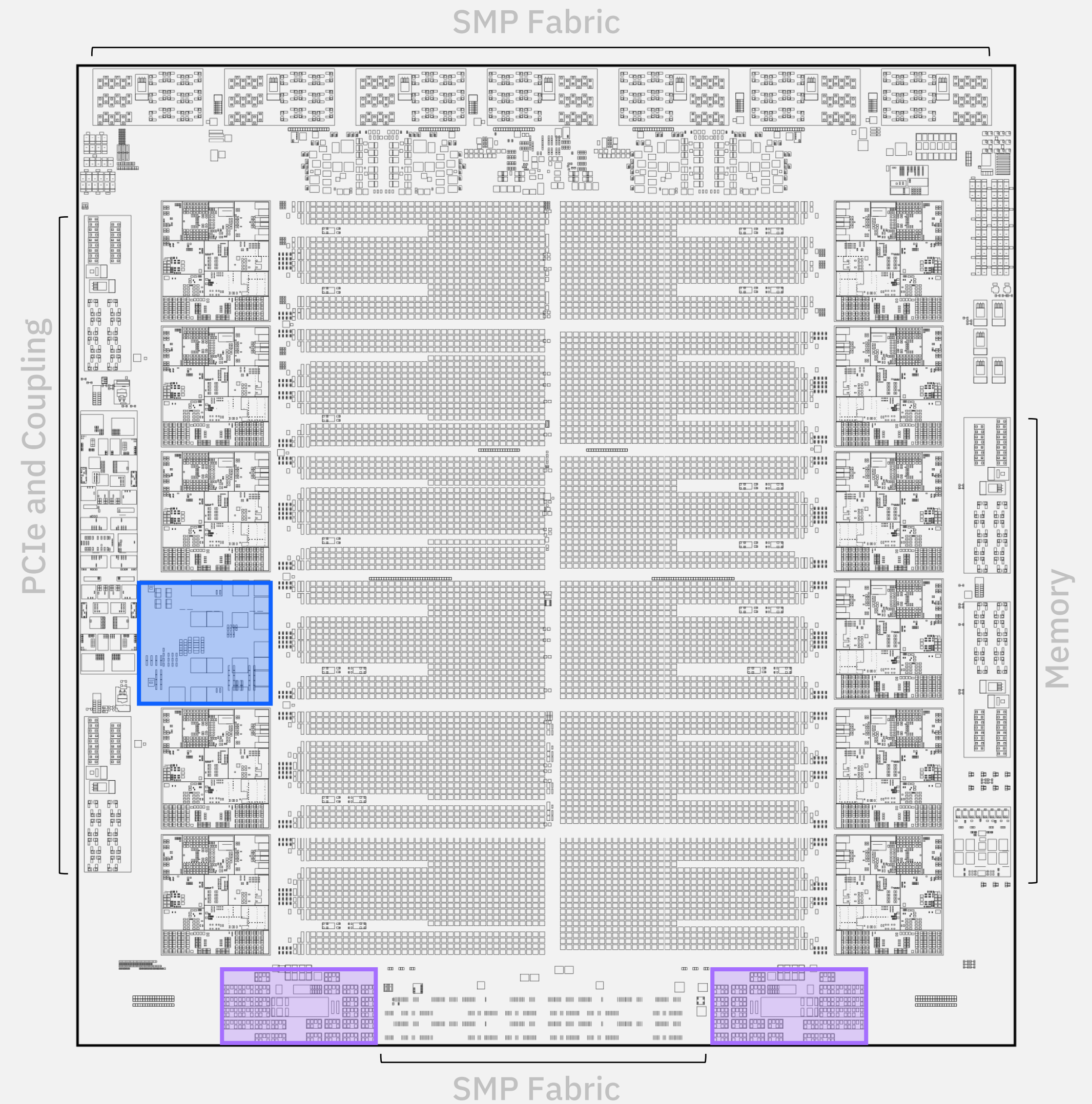
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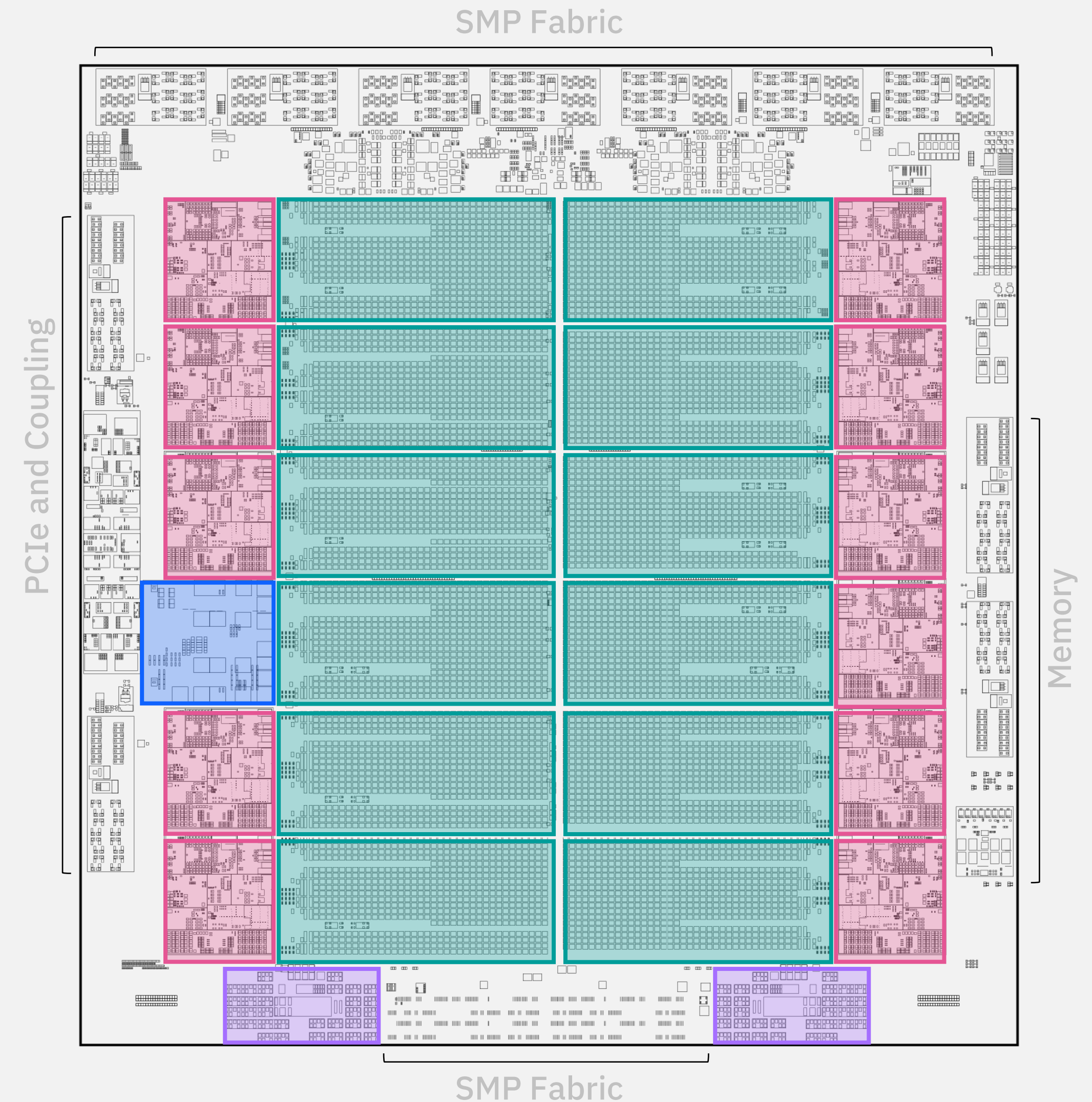
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# AArch64 implementation

## Compatibility

Full hardware implementation of AArch64 v9.3

Scalable Vector Extension (SVE, SVE2) support

Arm SystemReady

## Architecture

CISC Mainframe DNA

- High Frequency
- Large BTB, TLBs and Caches
- Reliability
- Strong ordering

## Implementation highlights

Little-Endian Arm execution in Big-Endian micro-architecture

2,792(!) AArch64 Instructions

239 AArch64 System Registers

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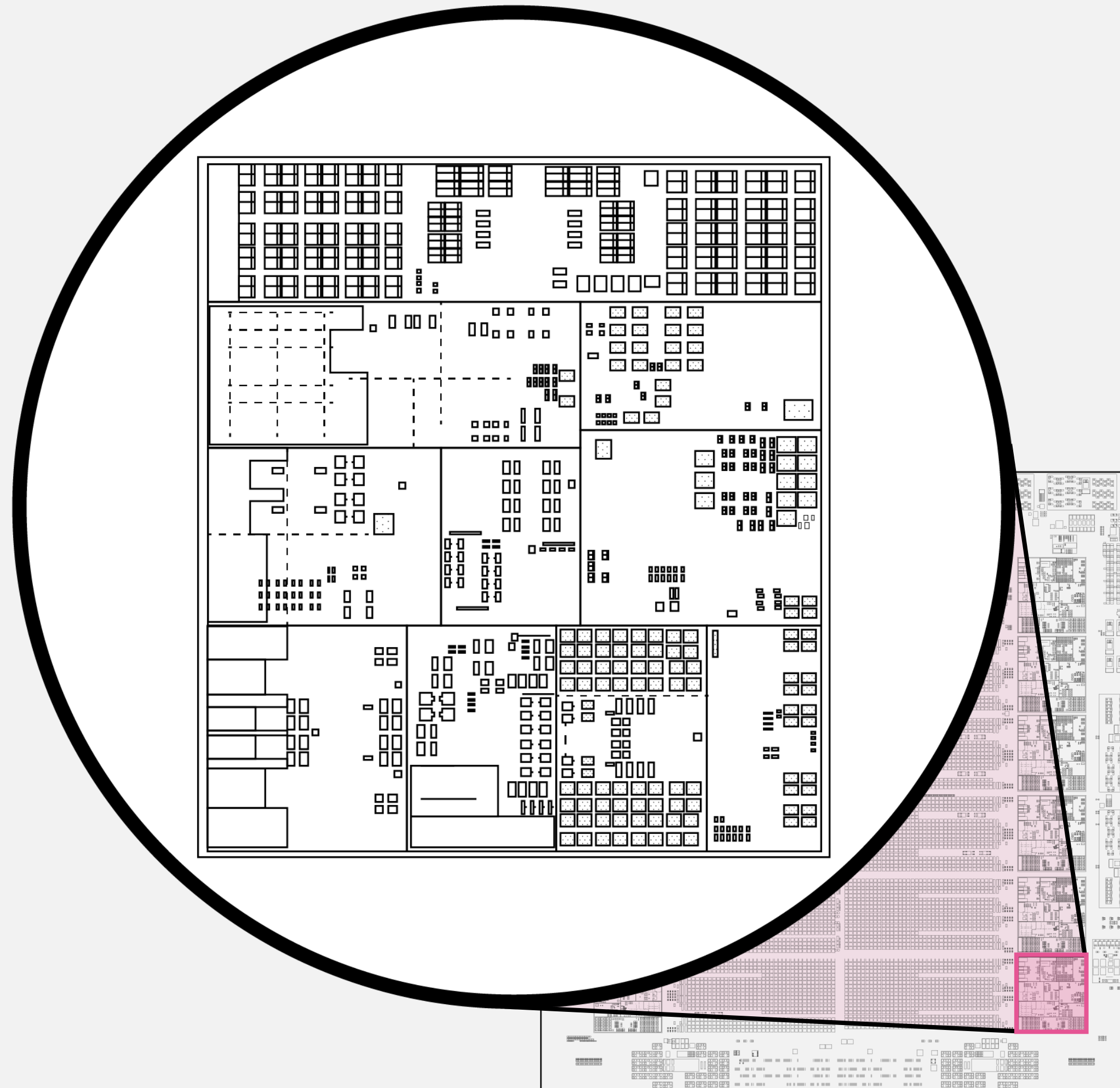
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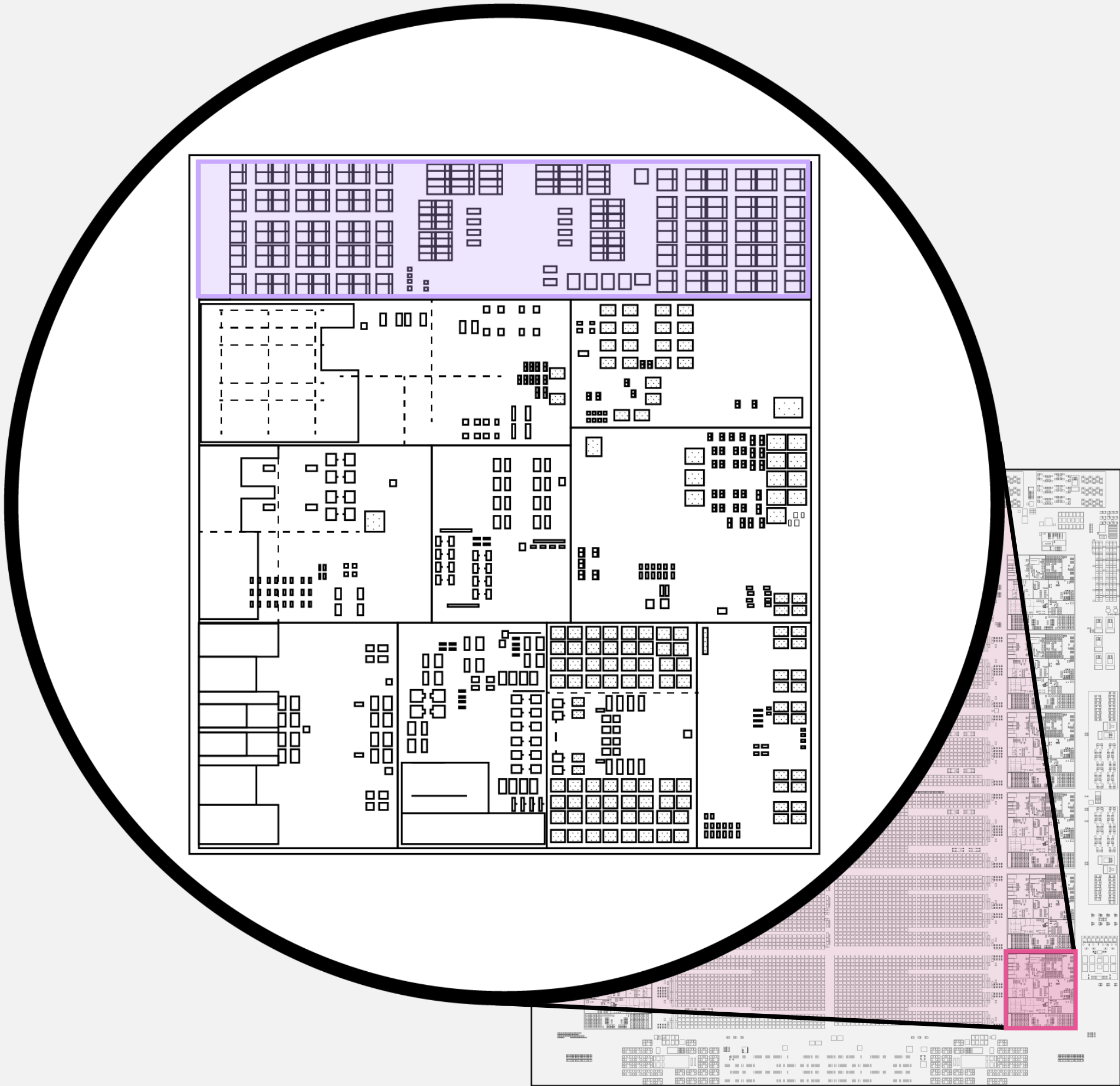
# AArch64 changes to the core



# AArch64 changes to the core

 **Branch Prediction**

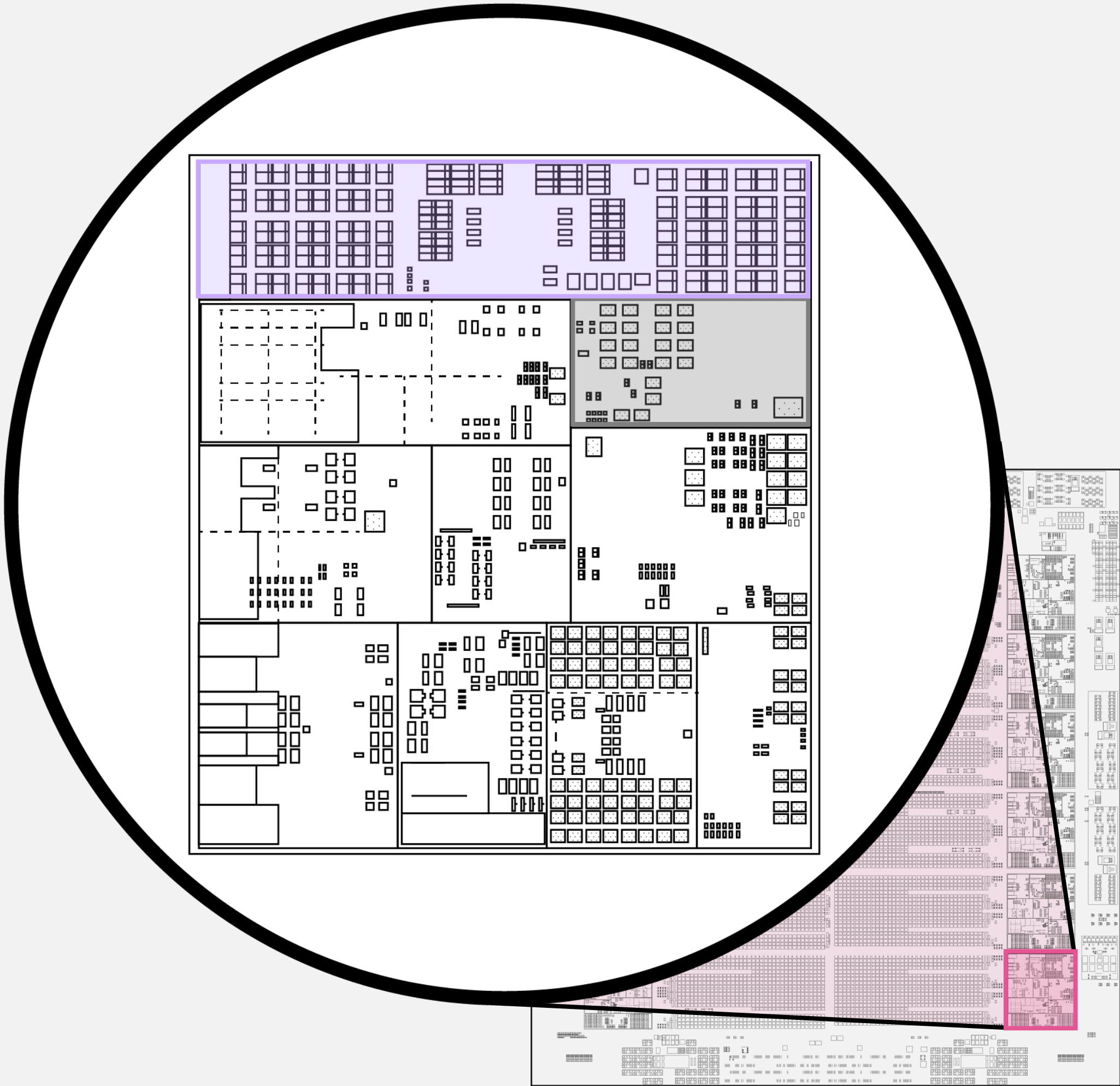
- Leverage existing design



# AArch64 changes to the core

- Fetch
  - Virtual cache tags

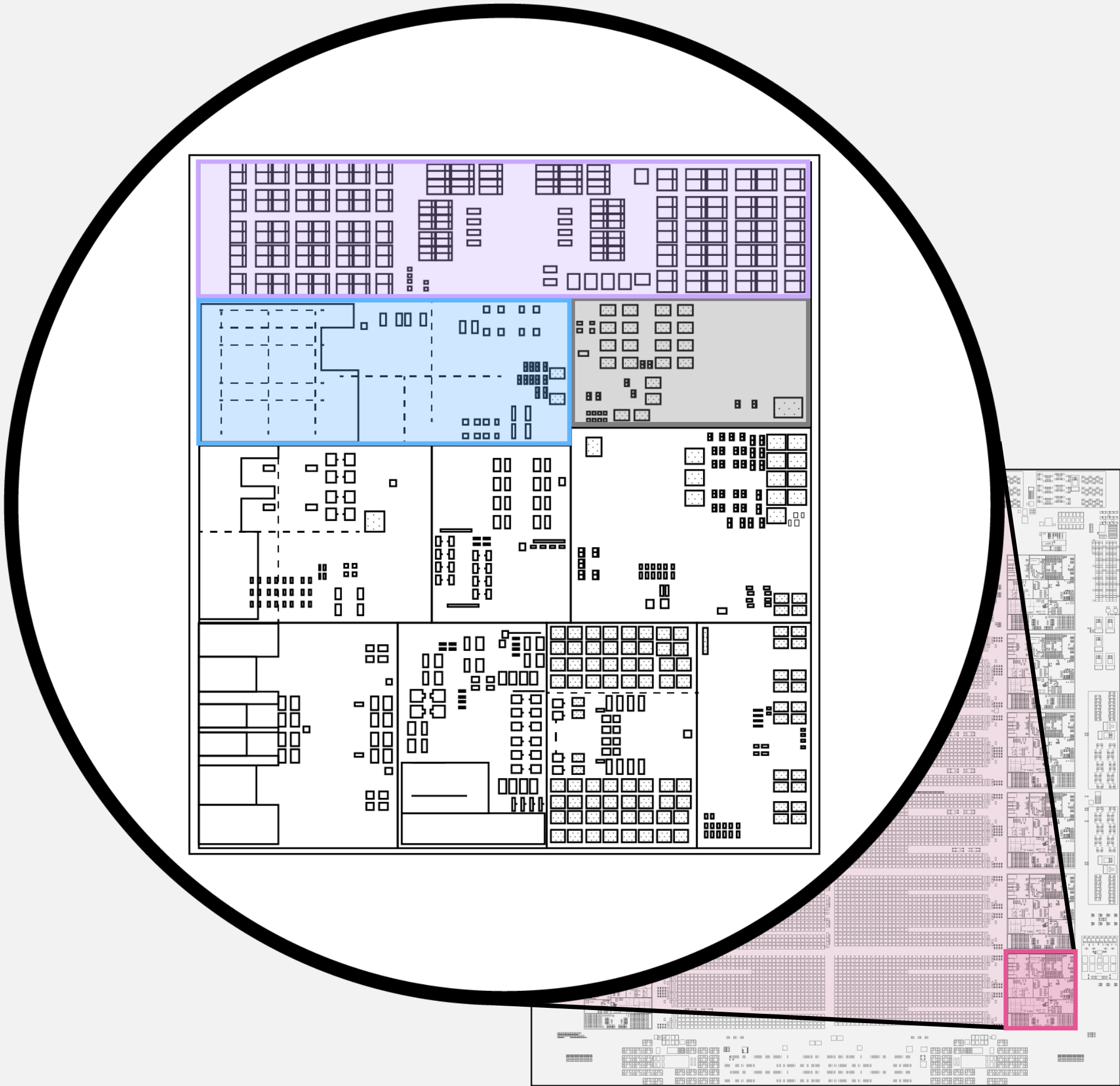
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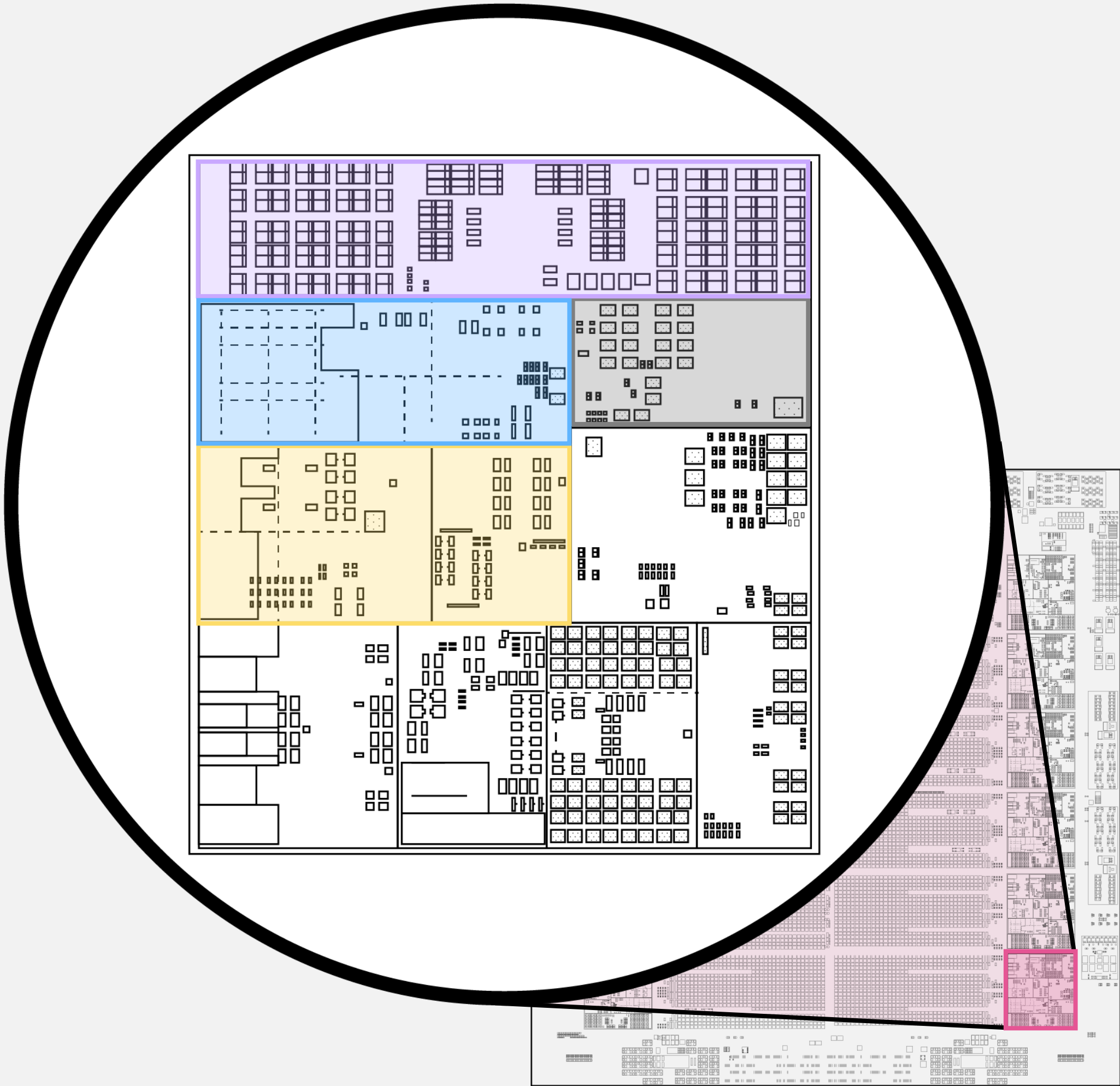
- Decode**
  - Automation consumes ARM XML

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# AArch64 changes to the core

- Dispatch and Issue**
  - Repurposed Register Rename for GR16-31

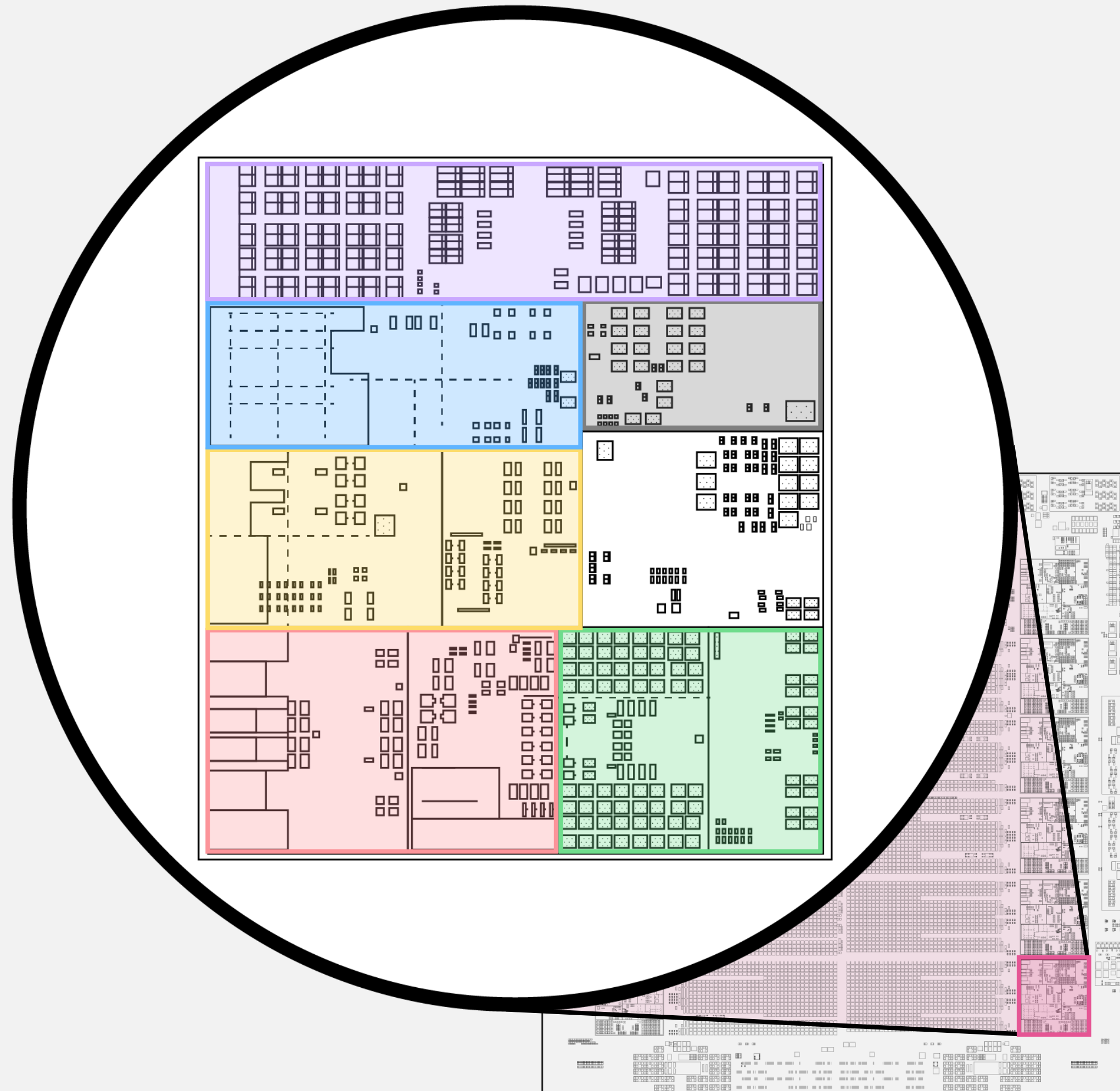


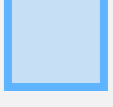
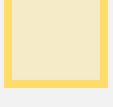
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# AArch64 changes to the core

## Arithmetic and Load-Store Store

- Reuse of major dataflows
- New control for SVE
- New dataflow: FP16/Bfloat16, Crypto
- Non-obvious CISC reuse (e.g. memory copy and clear)

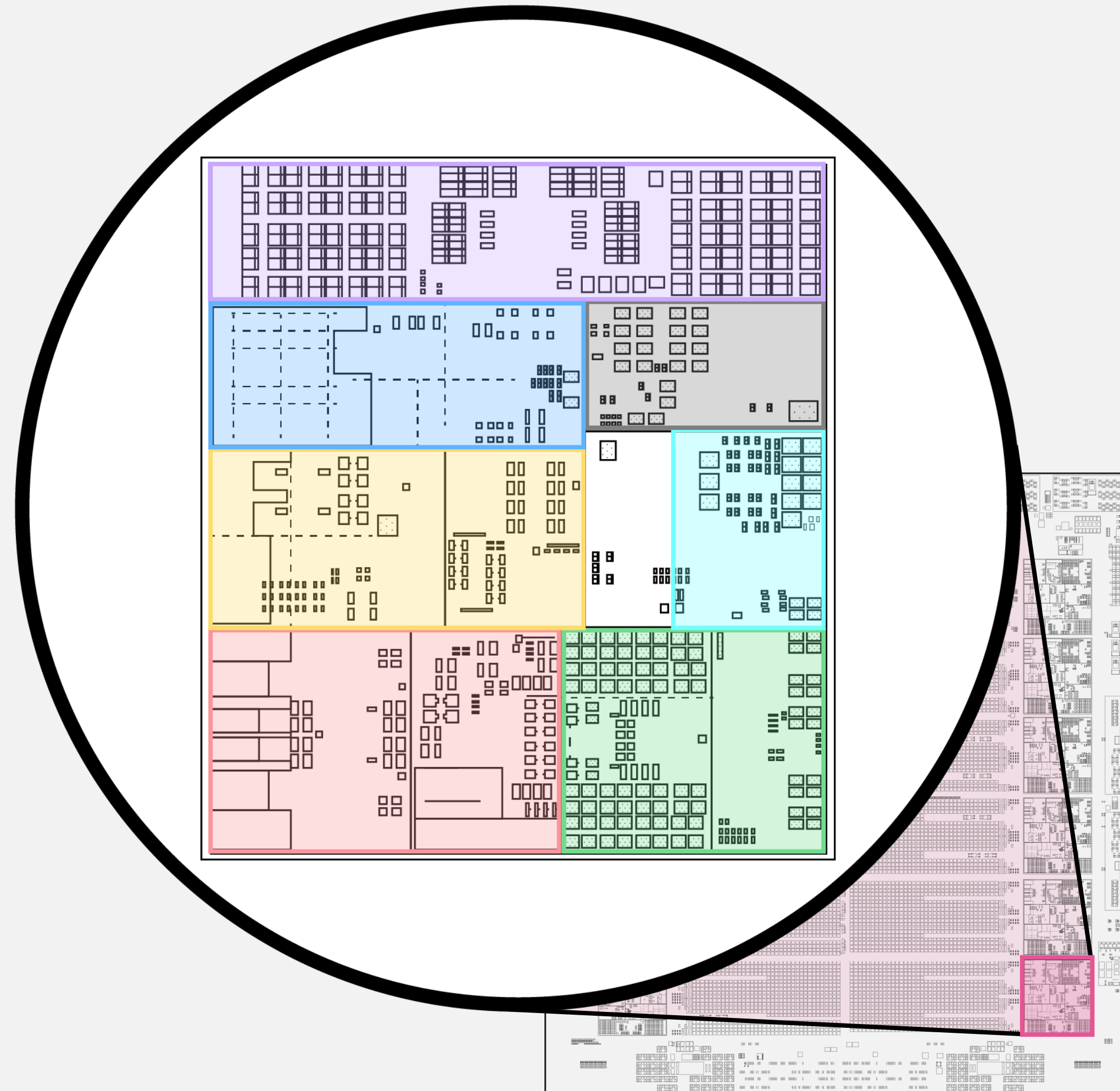


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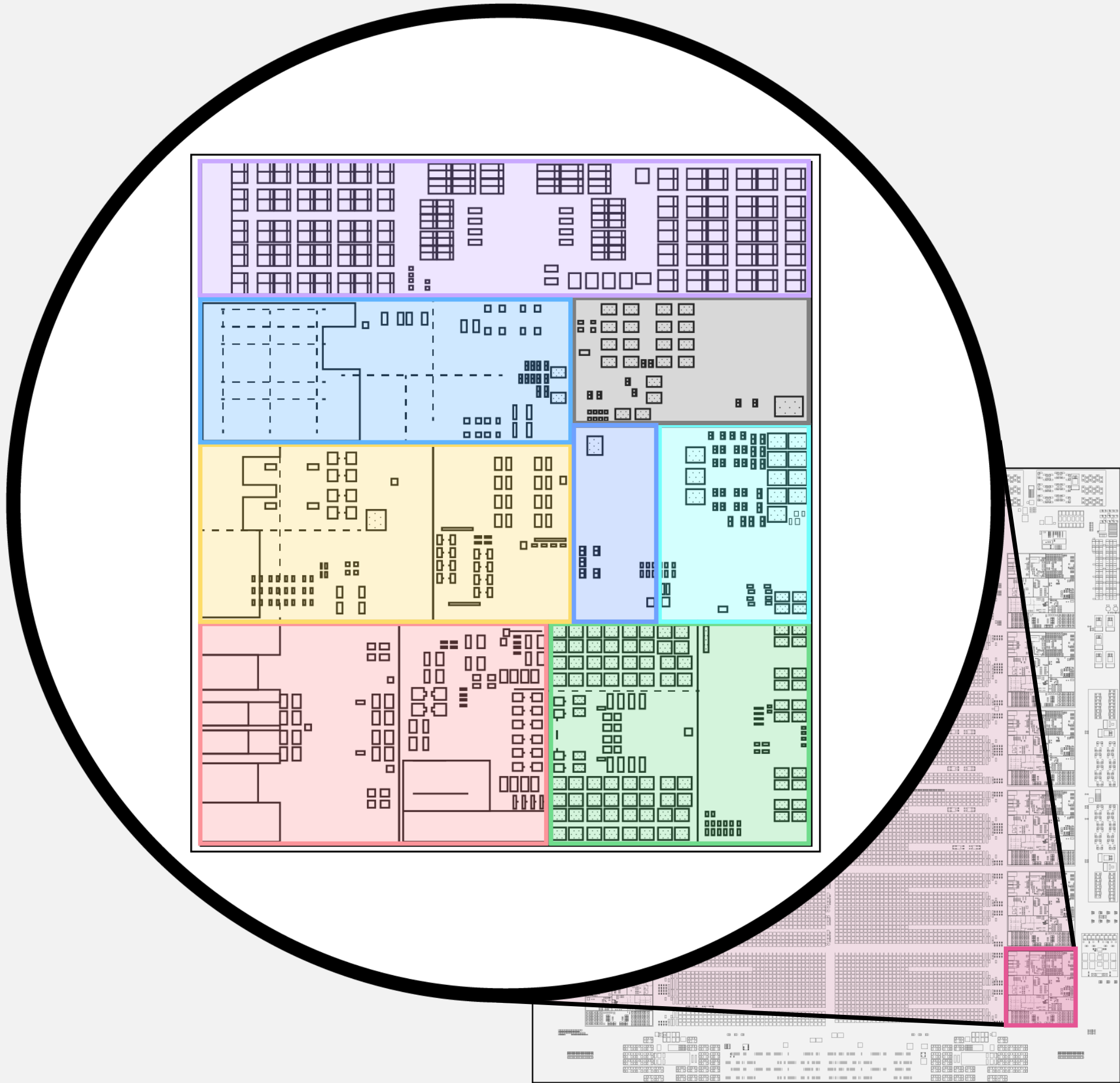
- Reused TLB
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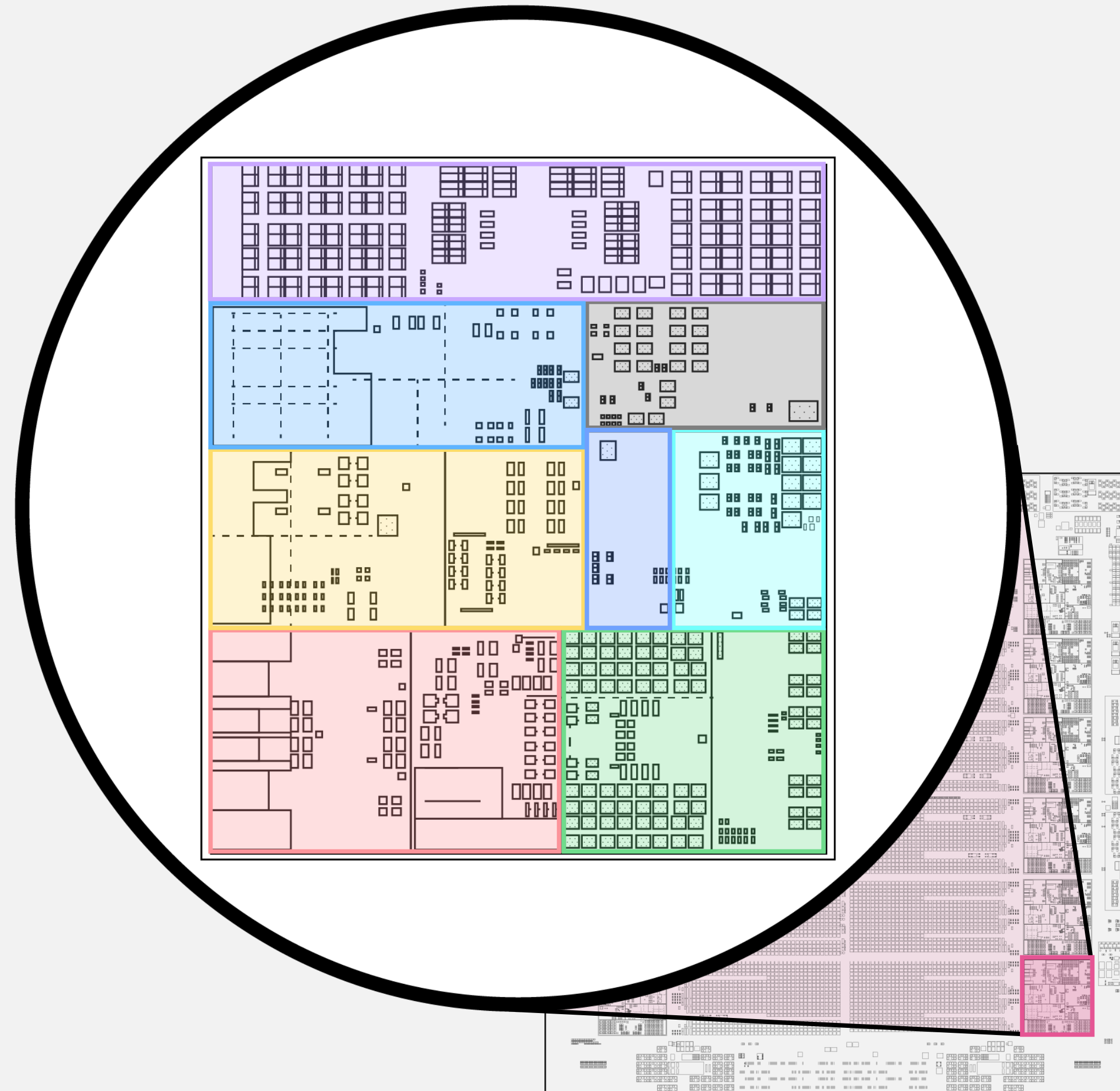
# AArch64 changes to the core

- Recovery Unit**
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# Arm software meets Z acceleration

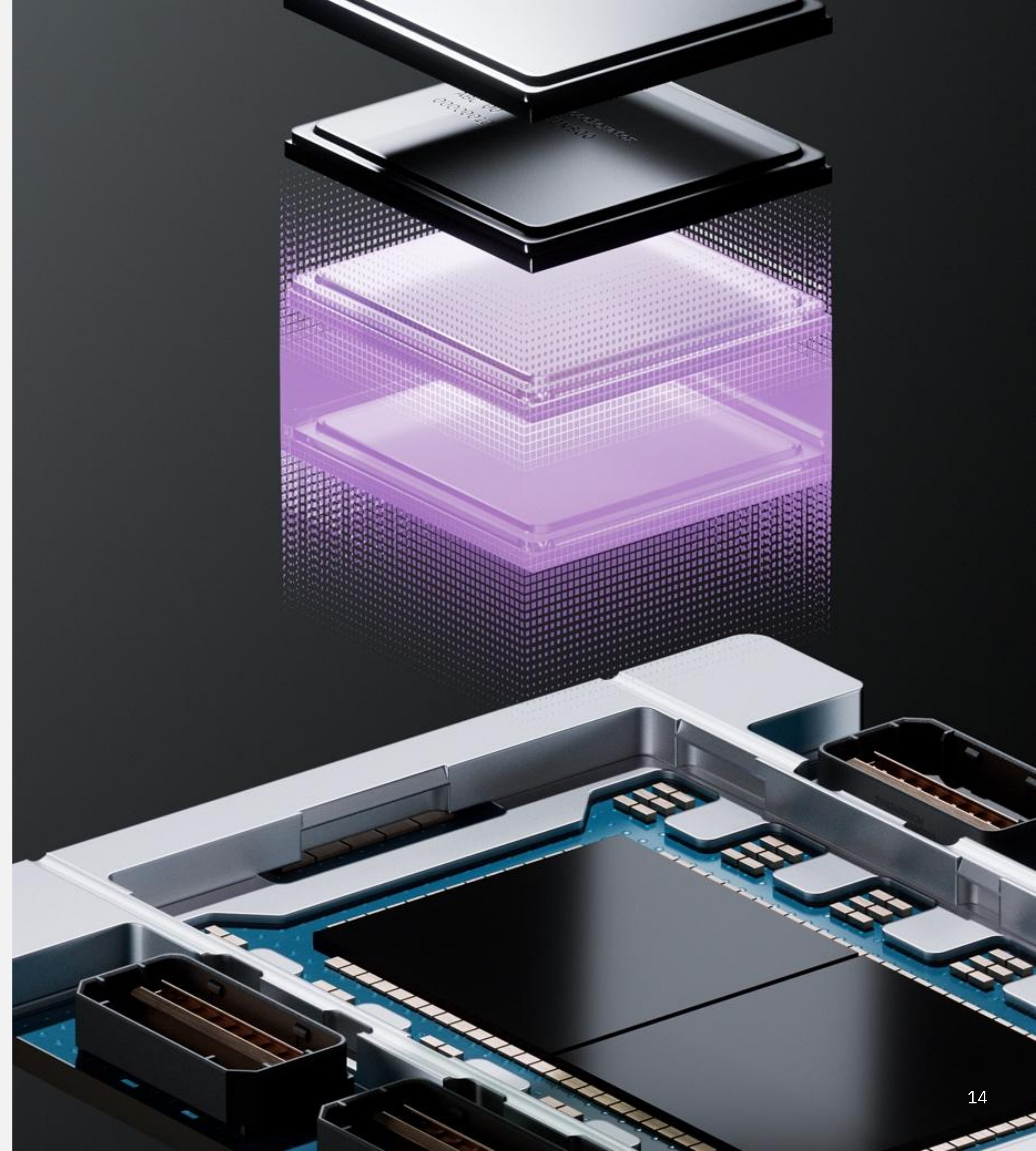
## Z accelerator optimization for transaction use cases

- Crypto
- GZIP
- On-Chip AI  
Acceleration

## Z ISA exposes them as instructions

## Exposed as Linux platform devices on Arm

- Full memory-  
management support
- Comparable  
latency to Z



99.99999999

availability  
%

# 99.9999999%

- Error checking across arrays, dataflows, and control
- Transparent recovery from transient faults
- Core sparing for persistent faults
- Concurrent repair
- Redundant Array of Independent Memory (RAIM)



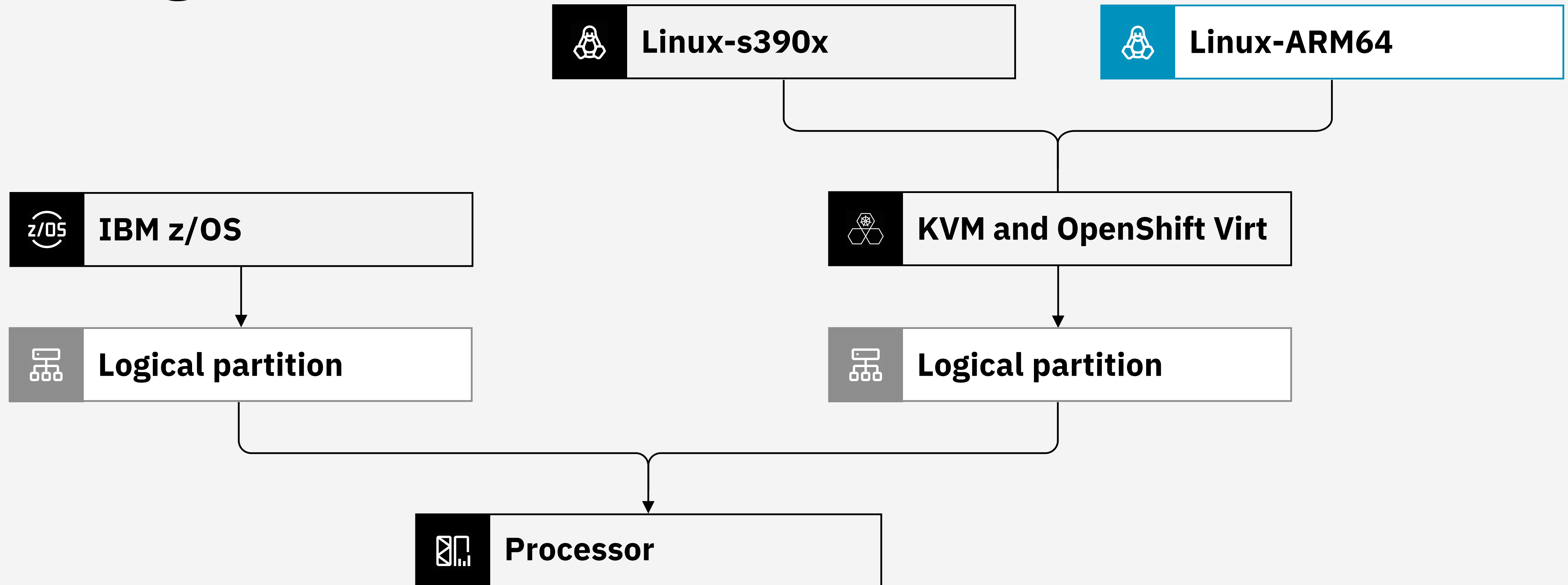
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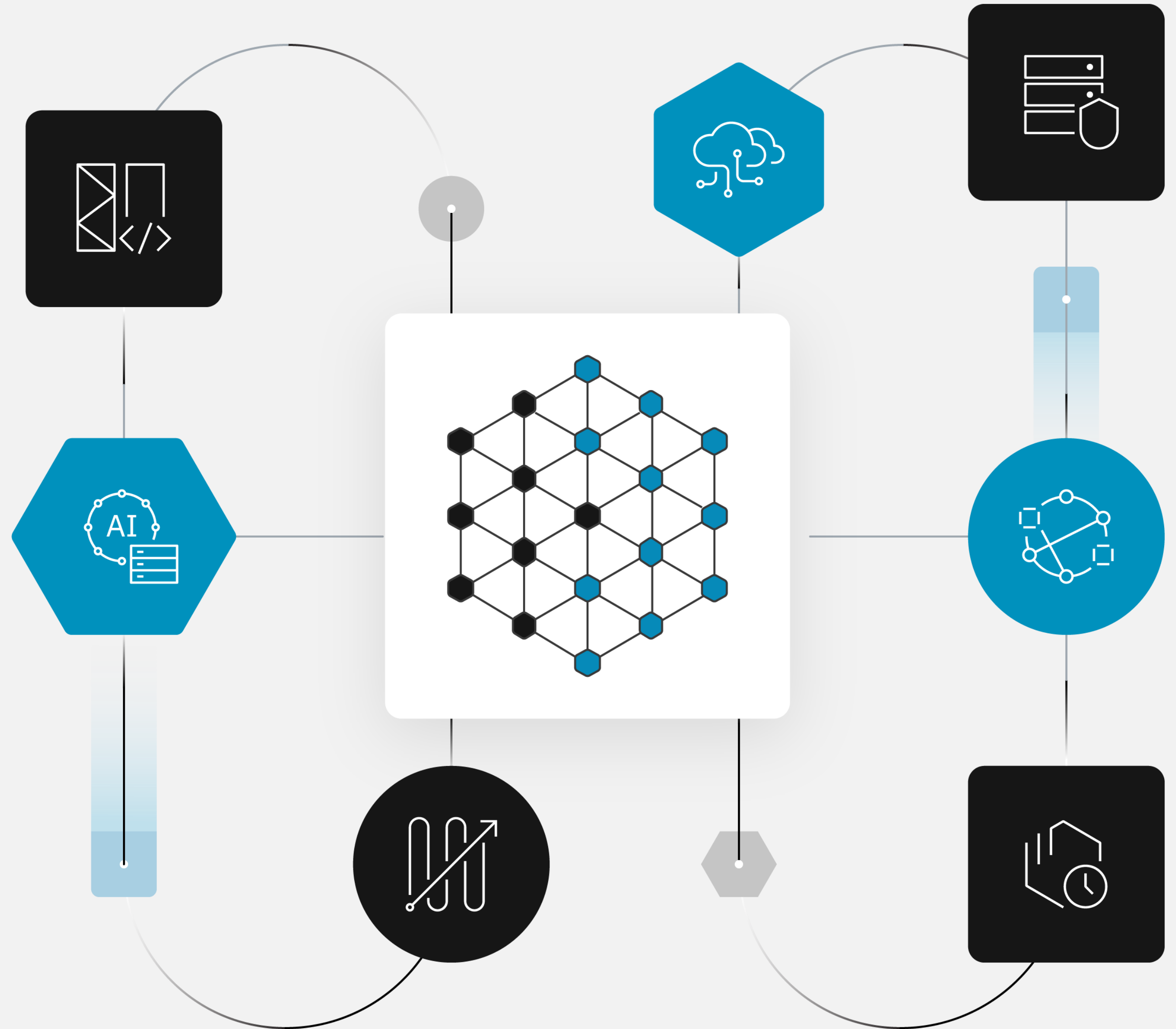
# IBM Z and Arm

## Coexisting through Linux KVM

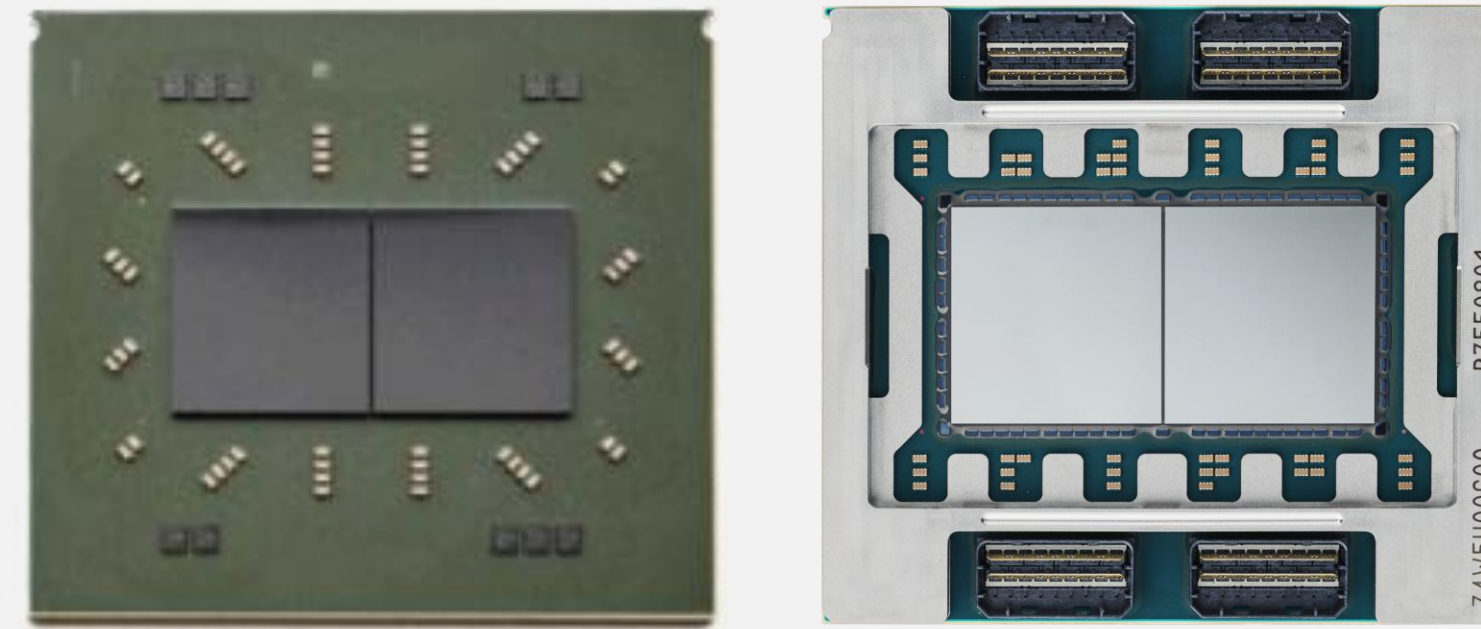


# IBM Z and Arm

All the benefits  
of both software  
ecosystems



# AI Acceleration

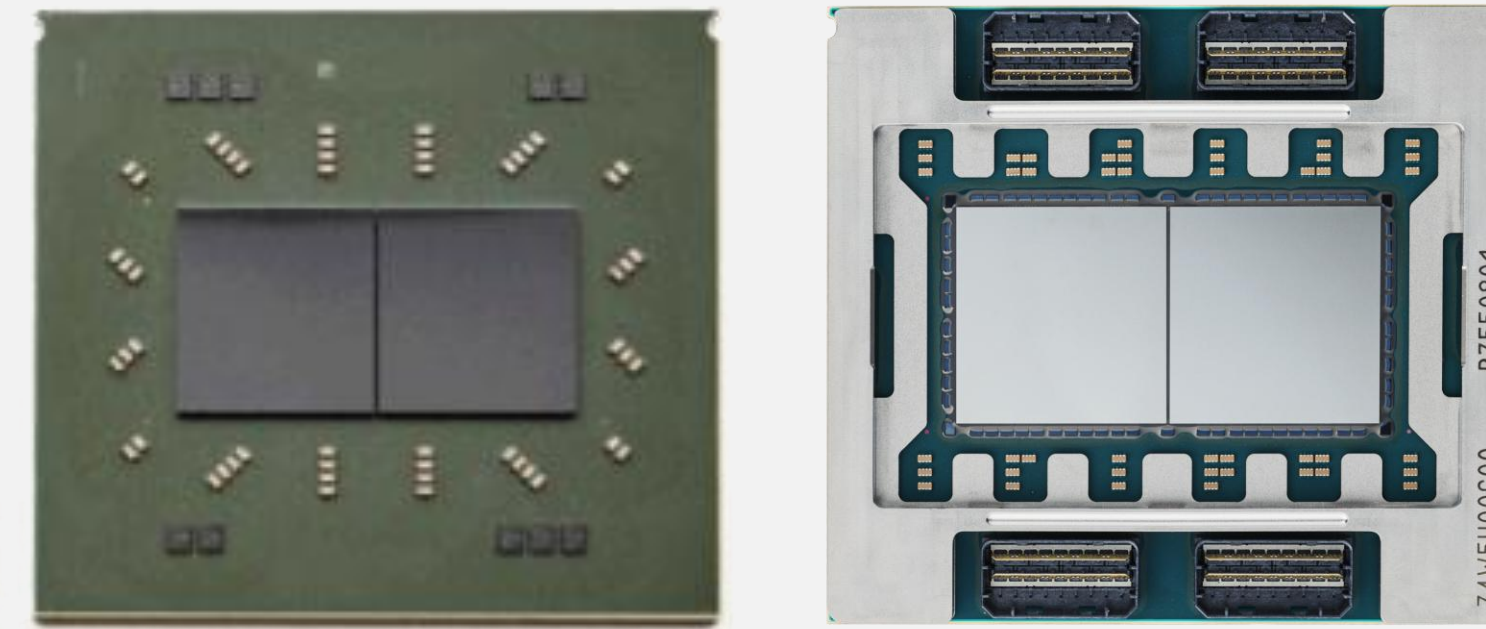


**On-Processor (Telum I & II)**



**Spyre PCIe Card**

# AI Acceleration



On-Processor (Telum I & II)



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## Enterprise AI use cases

Business processes

AI Operations

Agentic software  
development

# Enterprise GenAI inference acceleration

## Designed for larger enterprise inference workloads

Power envelope and performance vary by configuration

### AI cores

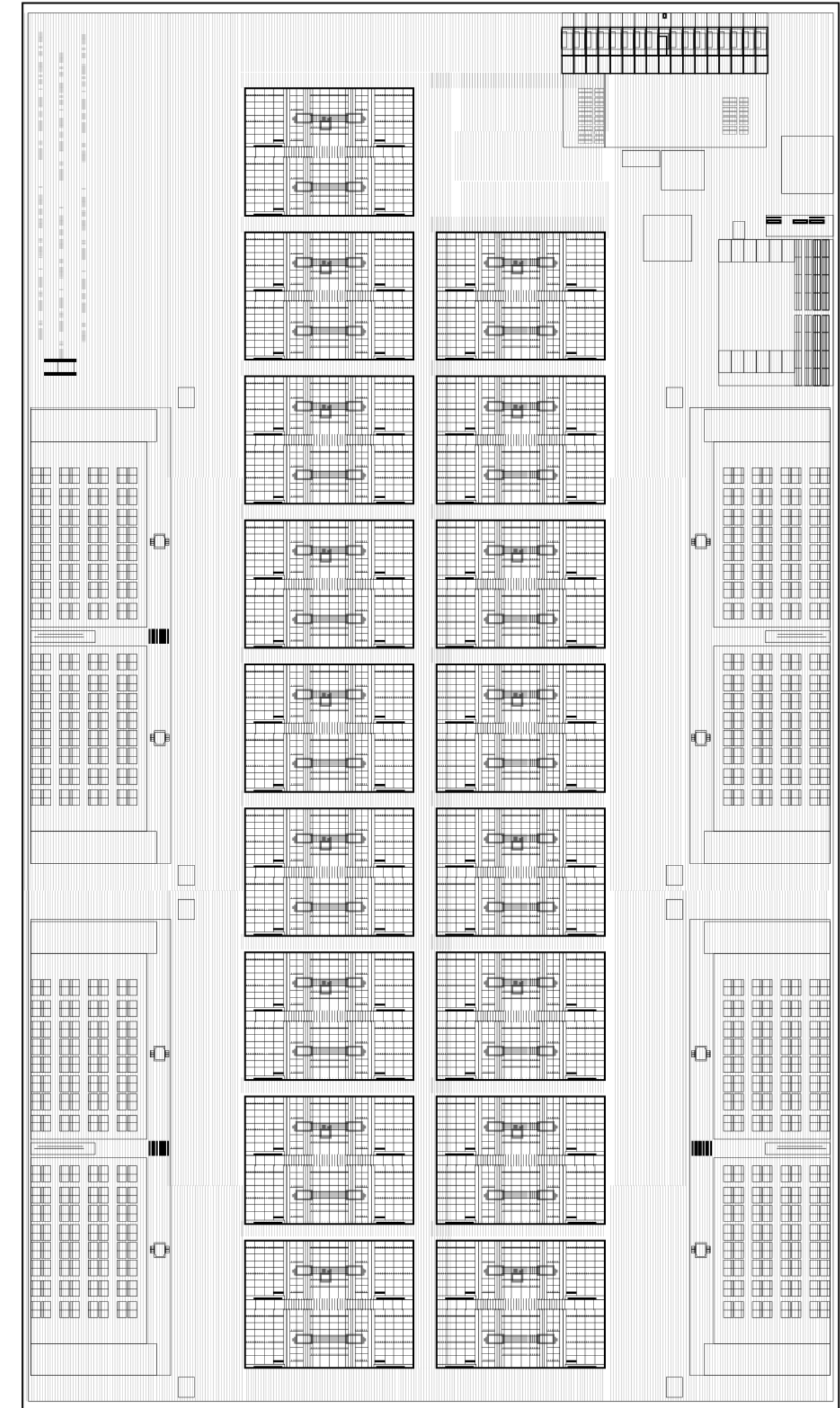
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- AI core changes for performance and FP4/MXFP4
- 600+ fp8 TOPS, 4x vs current generation

### HBM3e Memory

- 96GB HBM3e memory, ~4TB/s
- 20x vs current generation

### PCIe Gen6

- PCIe Gen6 x16
- Low-latency peer-to-peer interface



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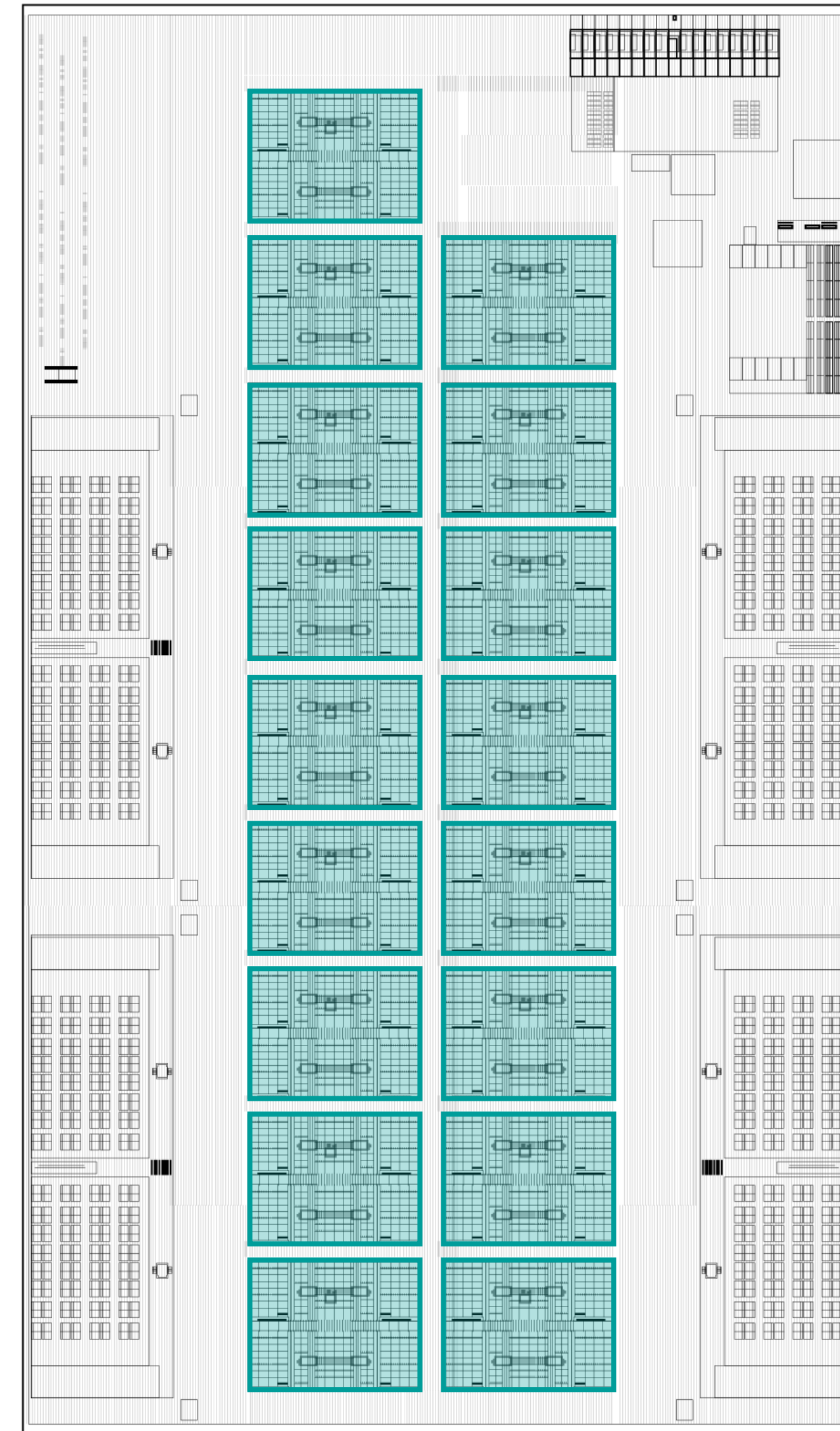
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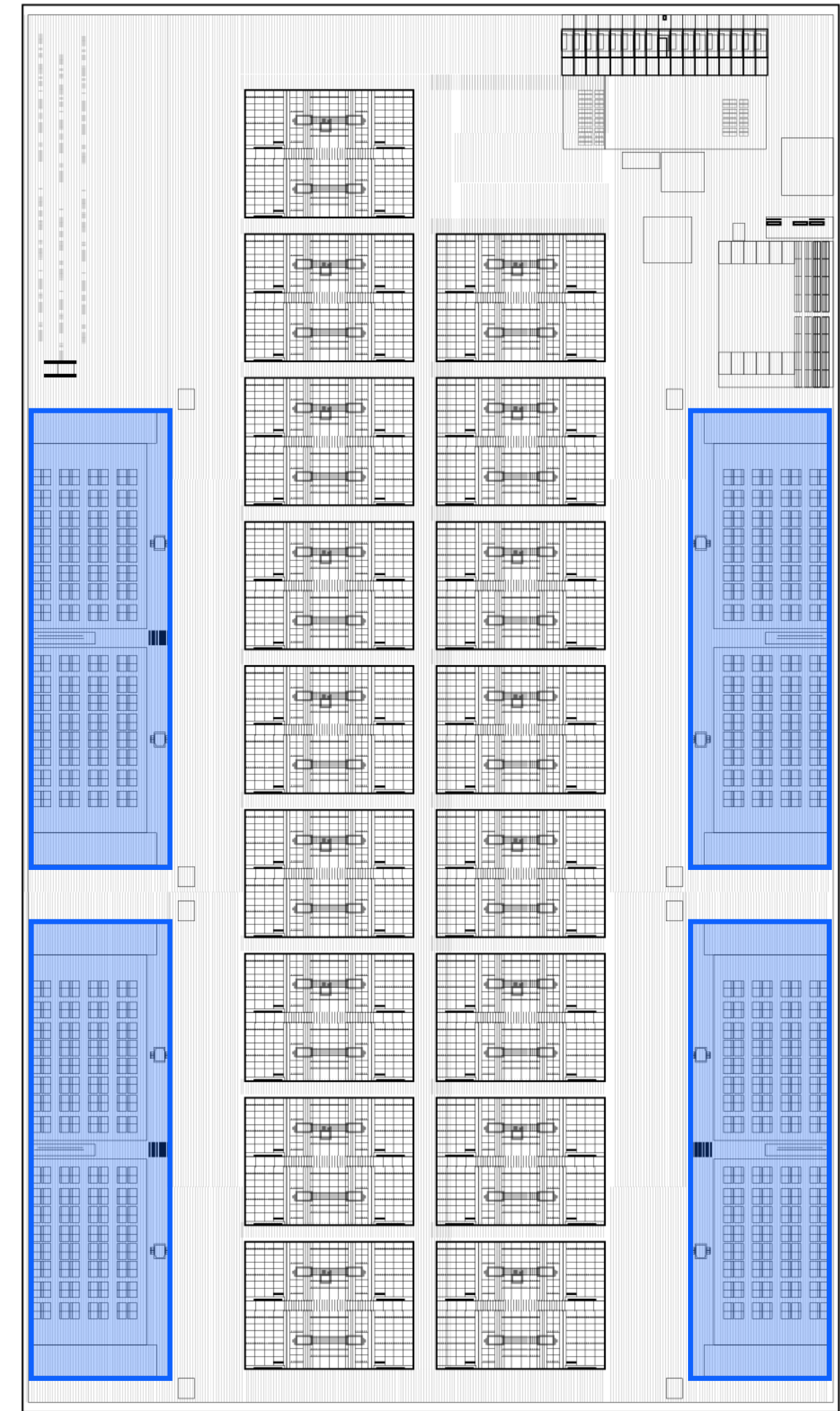
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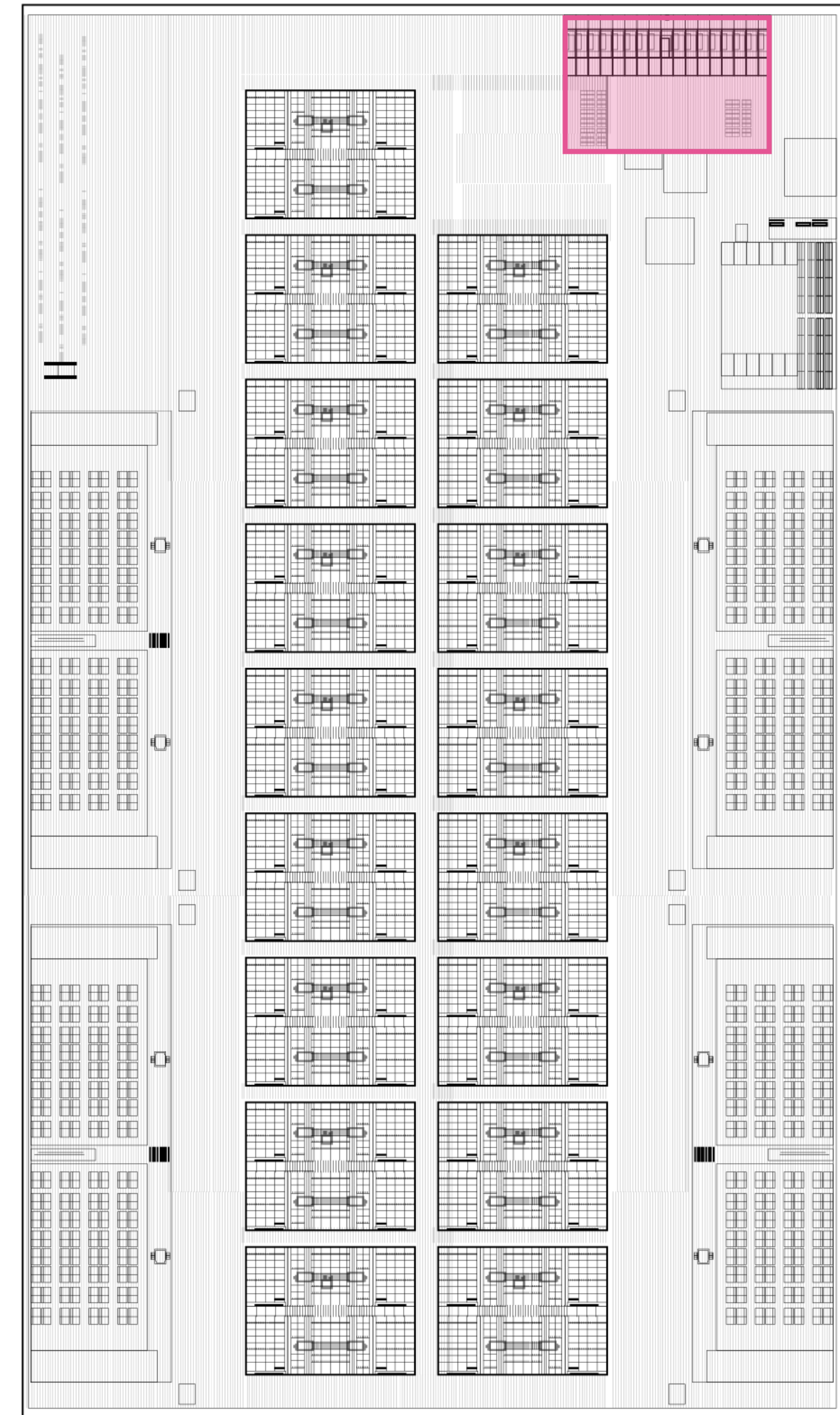
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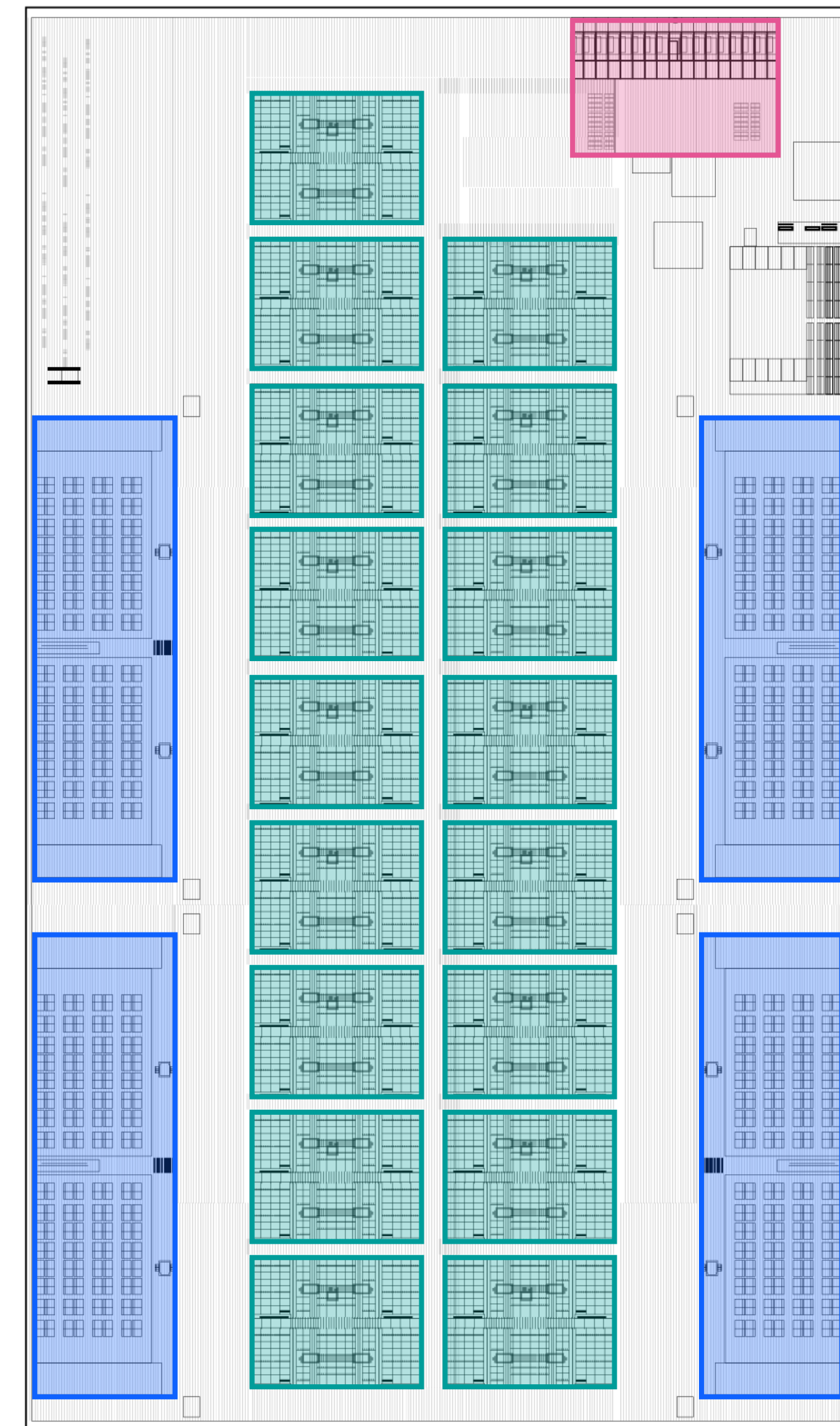
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# Secure and resilient AI

Designed for mission-critical AI workflows

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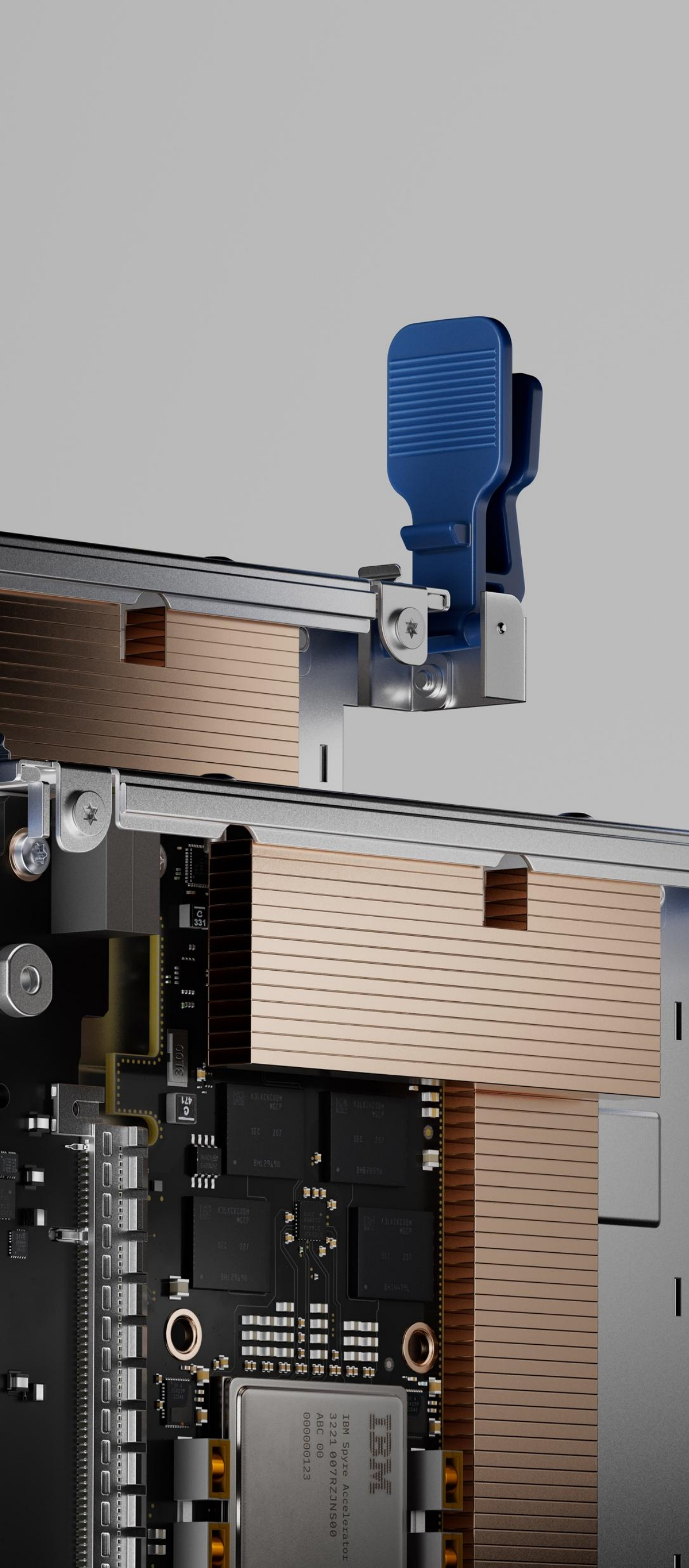
- Data, models and harness
- At rest, in transit, in use

## Features

- Confidential Computing with secure passthrough
- Quantum-Safe Secure boot
- On-Chip Cryptography

## RAS

- Firmware stack optimized for availability and serviceability
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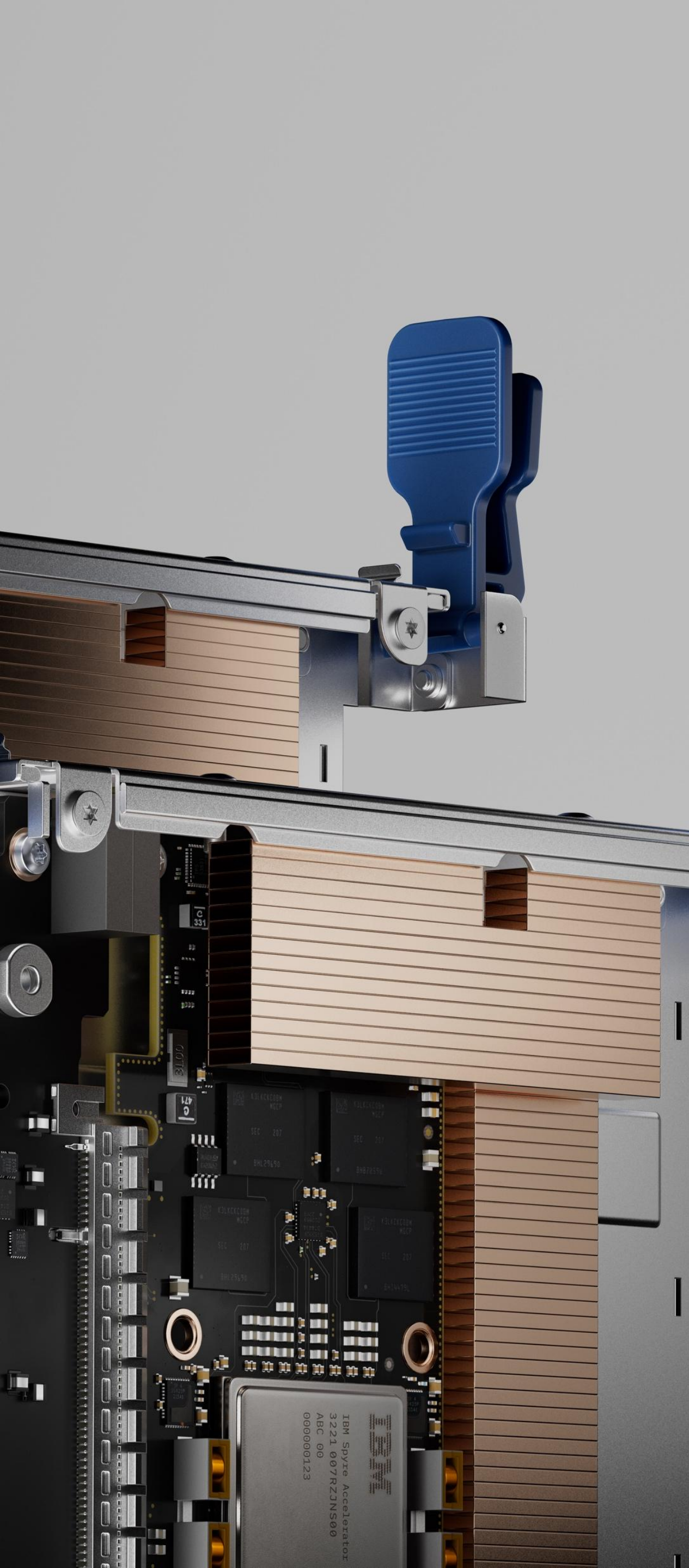
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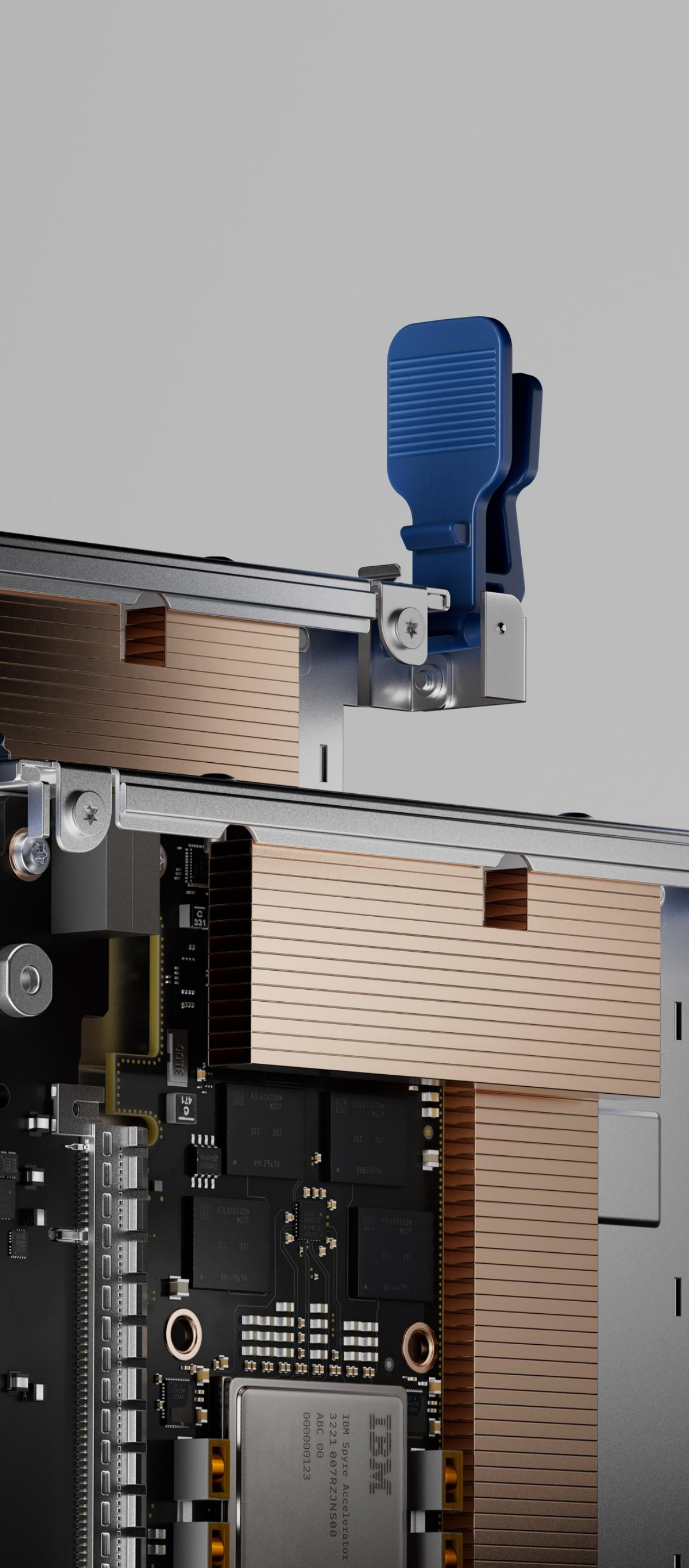
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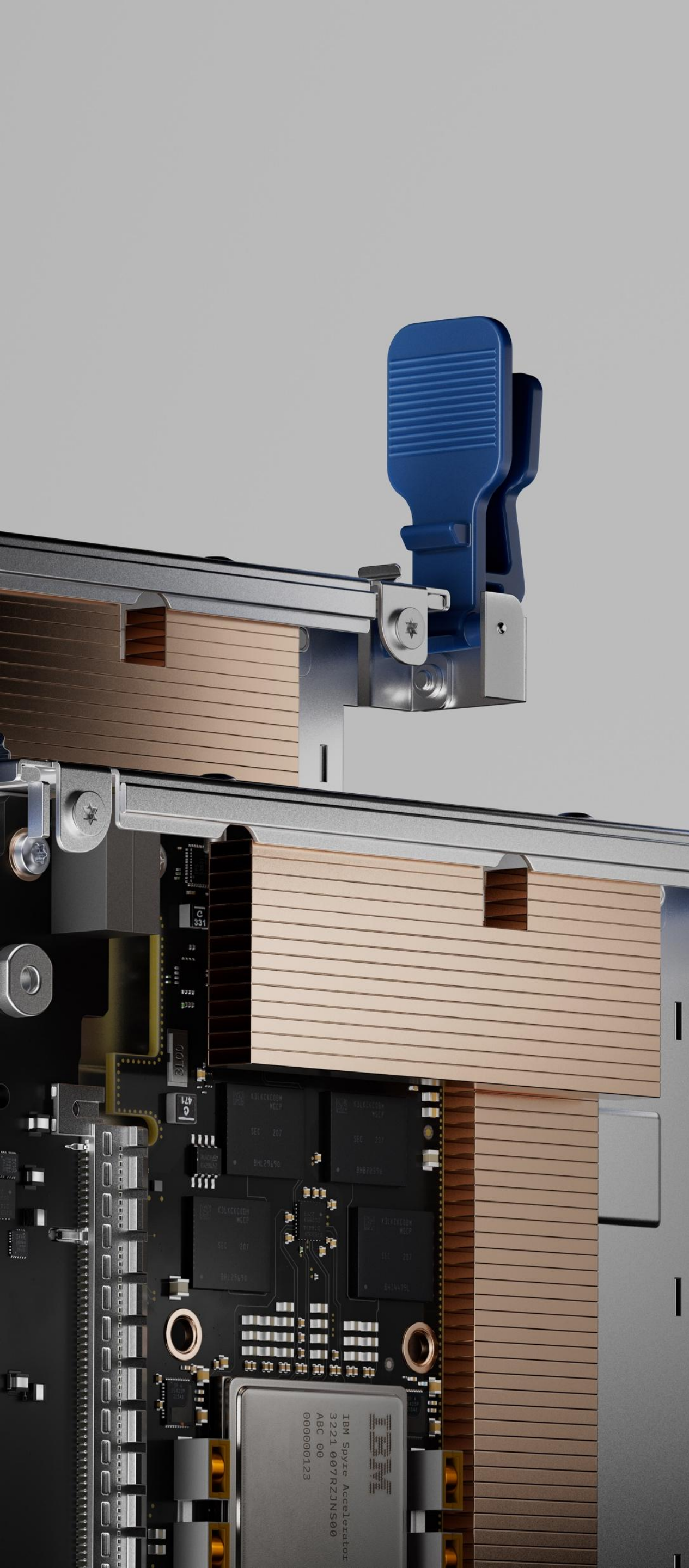
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Built for resilience  
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